Advanced Research Computing at Cardiff University

ARC at Cardiff University

1st October 2026



Jose Munoz

Advanced Research Computing @ Cardiff (ARCCA)

Training Opportunities and Schedule

COURSE : OCTOBER – DECEMBER 2026	TIMES AND DATES
1. ARC: Advanced Research Computing at Cardiff University	Thu 1st October 2026, 1:00 pm – 3:00 pm
2. ARC: An Introduction to Linux with Command Line (& Windows)	Thu 8th October 2026, 1:00 pm – 5:00 pm
3. ARC: An Introduction to Linux Shell Scripting	Thu 15th October 2026, 1:00 pm – 5:00 pm
4. ARC: Supercomputing for Beginners	Thu 22nd October 2026, 1:00 pm – 5:00 pm
5. ARC: Supercomputing – Beyond the basics	Thu 5th November 2026, 1:00 pm – 5:00 pm

Current training schedule; please check

https://arcca.github.io

For help and questions: *arcca-help@cardiff.ac.uk*

Training Opportunities and Schedule

COURSE : OCTOBER – DECEMBER 2026	TIMES AND DATES
6. ARC: Introduction to Machine Learning Applications	Thu 12th November 2026, 1:00 pm – 5:00 pm
7. ARC: Introduction to Parallel Programming using OpenMP and MPI	Thu 19th November 2026, 1:00 pm – 5:00 pm
8. ARC: Introduction to Python	Thu 26th November 2026, 1:00 pm – 5:00 pm
9. ARC: Advanced Python	Thu 10th December 2026, 1:00 pm – 5:00 pm
10: ARC: Version Control with Git	Thu 17th December 2026, 1:00 pm – 5:00 pm

Current training schedule; please check

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Training Opportunities and Schedule

SPECIAL EVENTS	TIMES AND DATES
1. NVIDIA DLI - Fundamentals of Accelerated Computing with CUDA Python	Currently under review
2. Data Carpentry Workshop	Usually on late January

Current training schedule; please check

https://arcca.github.io

For help and questions: *arcca-help@cardiff.ac.uk*

Overview

- **The ARCCA team and ARCCA Services**
- **What is Advanced Research Computing?**
  - High Performance Computing – ARCCA
- **The UK HPC Landscape & Opportunities at Cardiff University?**
  - Resources available - Supercomputing Wales & the GW4 Tier-2 Facility
- **Why Advanced Research Computing?**
  - Background – Technology Trends; Rise of **AI & Machine / Deep Learning**
- **What is Parallel Computing?**
  - The Software Challenge and Scalability (Amdahl's Law)
- **Supercomputing at Cardiff**
  - **The Falcon Supercomputer** (Infrastructure), RSEs and Cloud Services.
  - Accessing Falcon & getting Started – User and Project Accounts and options for using the System (command line and OnDemand)
- **Survey (!!) and Concluding Remarks**

The ARCCA Team



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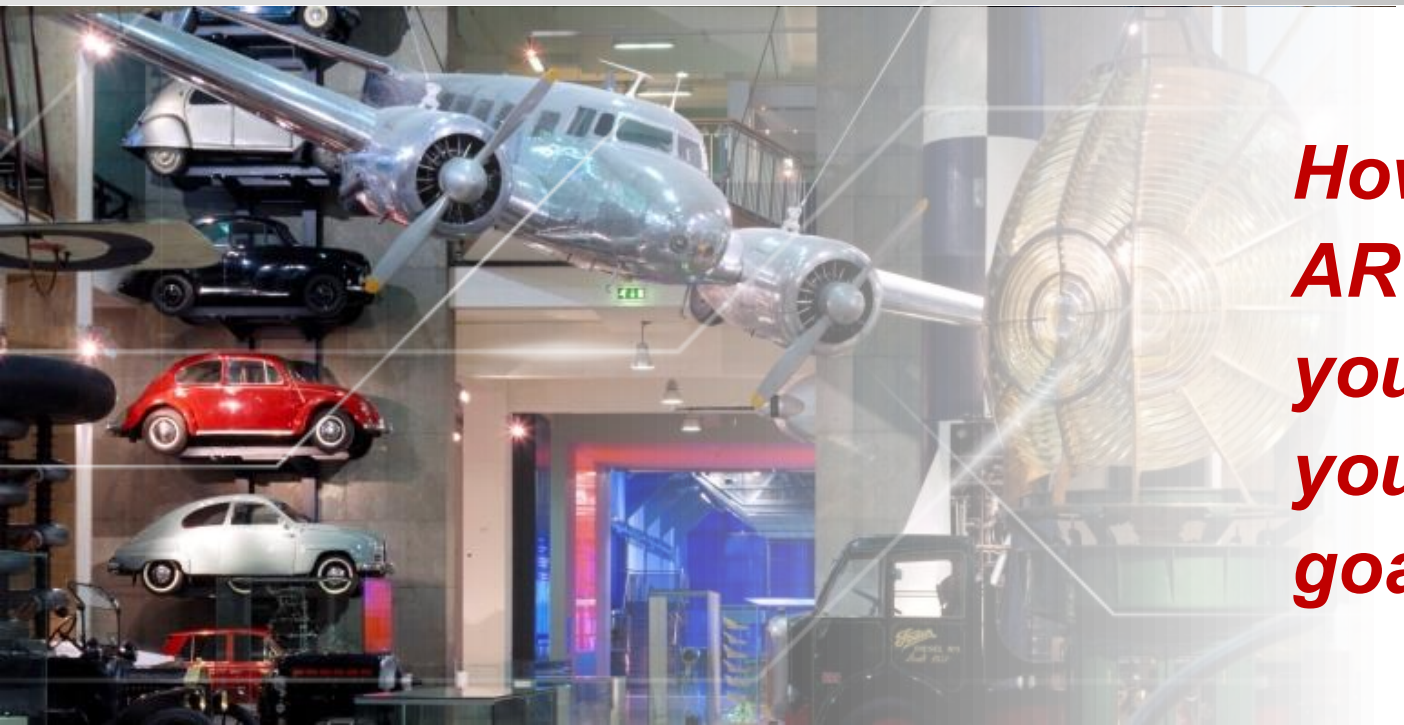
Dr Robyn Munoz
Research Software Engineer



Mr Connor Brock
Research Software Engineer

Mr Ieuan Underwood
Support Engineer (COMSC student placement)

Services provided by ARCCA

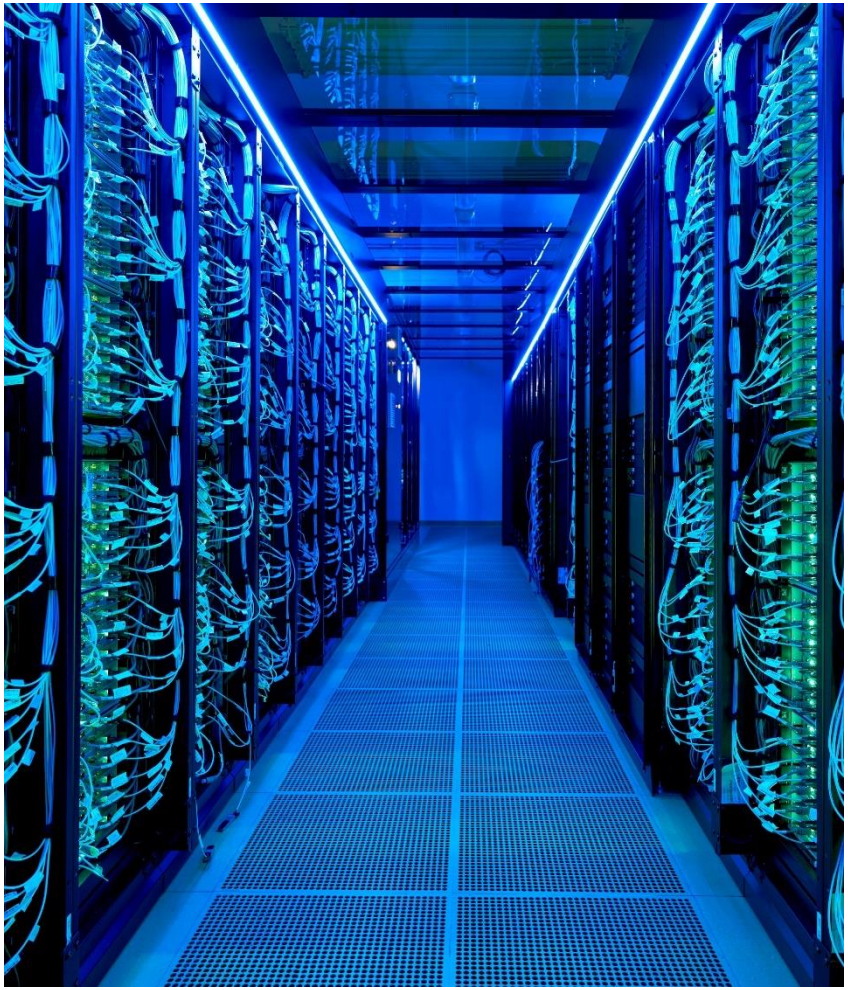


How can ARCCA help you accomplish your research goals.

ARCCA Services

Advanced Research Computing

- handling large quantities of statistical information
- manipulating and disseminating huge data-sets
- performing large numbers of calculations in **parallel**
- imaging and visualisation of large files
- large-scale or complex pattern matching and recognition.



ARCCA Services

Research Software Engineering (now available!)



- Software Development
- Data Engineering
- HPC Support
- Reproducibility
- Research Infrastructure

Research Data Store (available soon)



- provides secure, scalable storage for university students and researchers to manage large datasets and seamlessly process them using high-performance computing (HPC) resources.
- FAIR Data (Findable, Accessible, Interoperable, Reusable).

Cloud Services

- ARCCA carried out an initial pilot cloud project as part of wider University measures to explore research IT *options*.
- This demonstrated an interest from the research community in using cloud services.
- ARCCA is continuing with the service, with some changes to adapt to user requests, through **SPARKLE** as a discounted service provider for **AWS**. SPARKLE is billed by AWS.
- The service is available to funded projects.
- **Only** thing that is free is egress and ingress - **storage** and **compute** will contribute to costs.
- Get in touch if you require further details.

powered by aws

Research Software Engineers (RSEs)

- **Software is fundamental to research**. From the humanities to physics, biology to archaeology, software plays a vital role in generating results. Not all researchers can become skilled software engineers, so a **new role** has developed in academia: the **Research Software Engineer (RSE).**



- The term “Research Software Engineer” first coined in 2012. Since then, a joint campaign involving RSEs and the **Software Sustainability Institute** has led to the creation of the UK RSE Association, which in 2025 numbers over 700 paid members and 6900 members of their slack channel, the growth of Research Software Groups around the UK, the creation of a Fellowship sponsored by the EPSRC, the launch of the successful annual conference for RSEs and an increase in recognition for the role of the Research Software Engineer.

https://society-rse.org/

arcca Research Software Engineers (RSEs)

The new **ARCCA RSE** categorises five primary pillars:

Pillar	Service Description
Software Development	End-to-end creation of bespoke research tools, libraries (Python/R/C++), and web-based platforms for data analysis or visualisation.
Data Engineering	Designing robust data pipelines, ETL (Extract, Transform, Load) processes, and database schemas for large-scale or sensitive datasets.
HPC Support	Scaling local scripts to run on High-Performance Computing (HPC) clusters and GPU acceleration
Reproducibility	Implementation of "FAIR" software principles: version control, containerisation (Docker/Singularity), and automated testing.
Research Infrastructure	Building and maintaining shared platforms, such as data repositories, API gateways, or automated laboratory workflows.

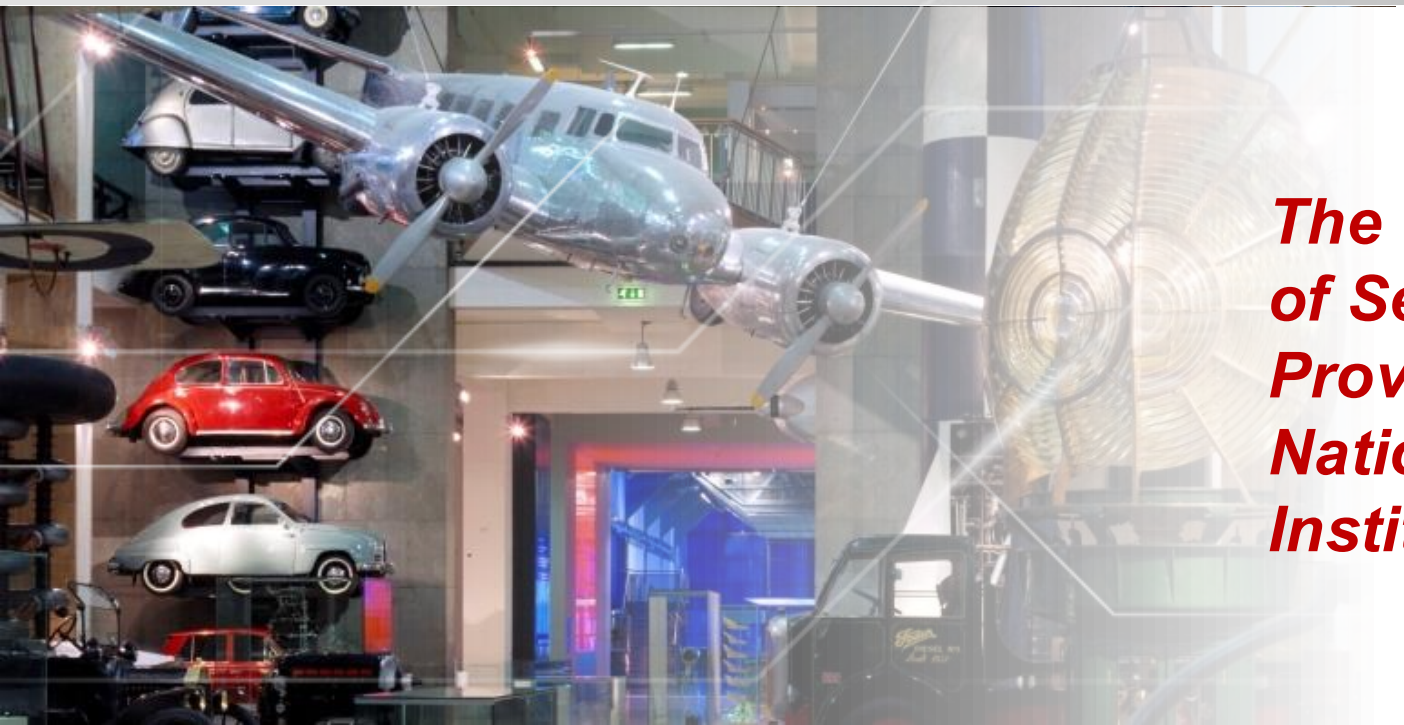
Research Software Engineers (RSEs)

- The new ARCCA RSE team is ready to help the Cardiff community!
- Please contact us at arcca@cardiff.ac.uk if you would like to know more how our RSEs can support your work or if you have a specific project where you think we can help you.

ARCCA RSE Service
(VPN required)

https://wiki.arcca.cf.ac.uk/index.php/The_Research_Software_Engineering_(RSE)_Service

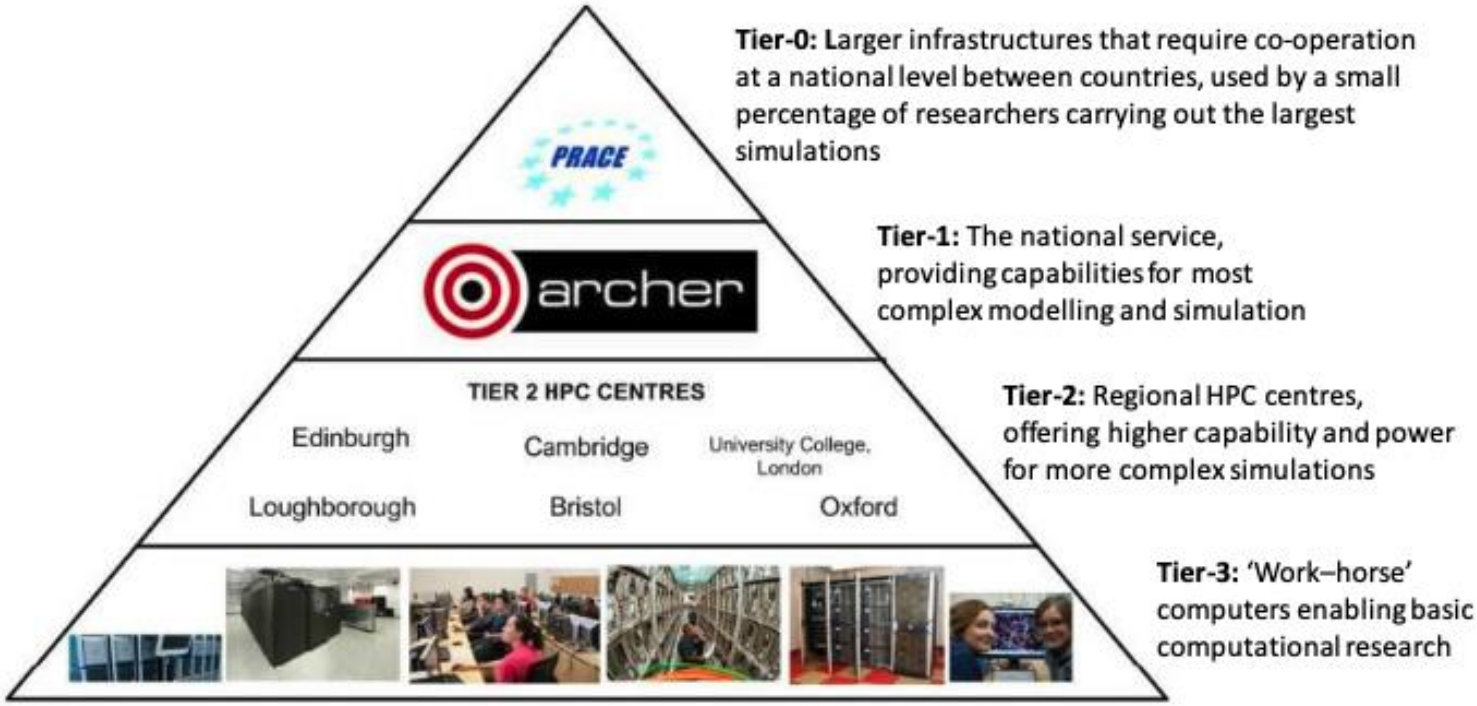
The UK HPC Landscape



The Tiered levels of Service Provision – from National to Institutional

Tier-2 HPC Centres

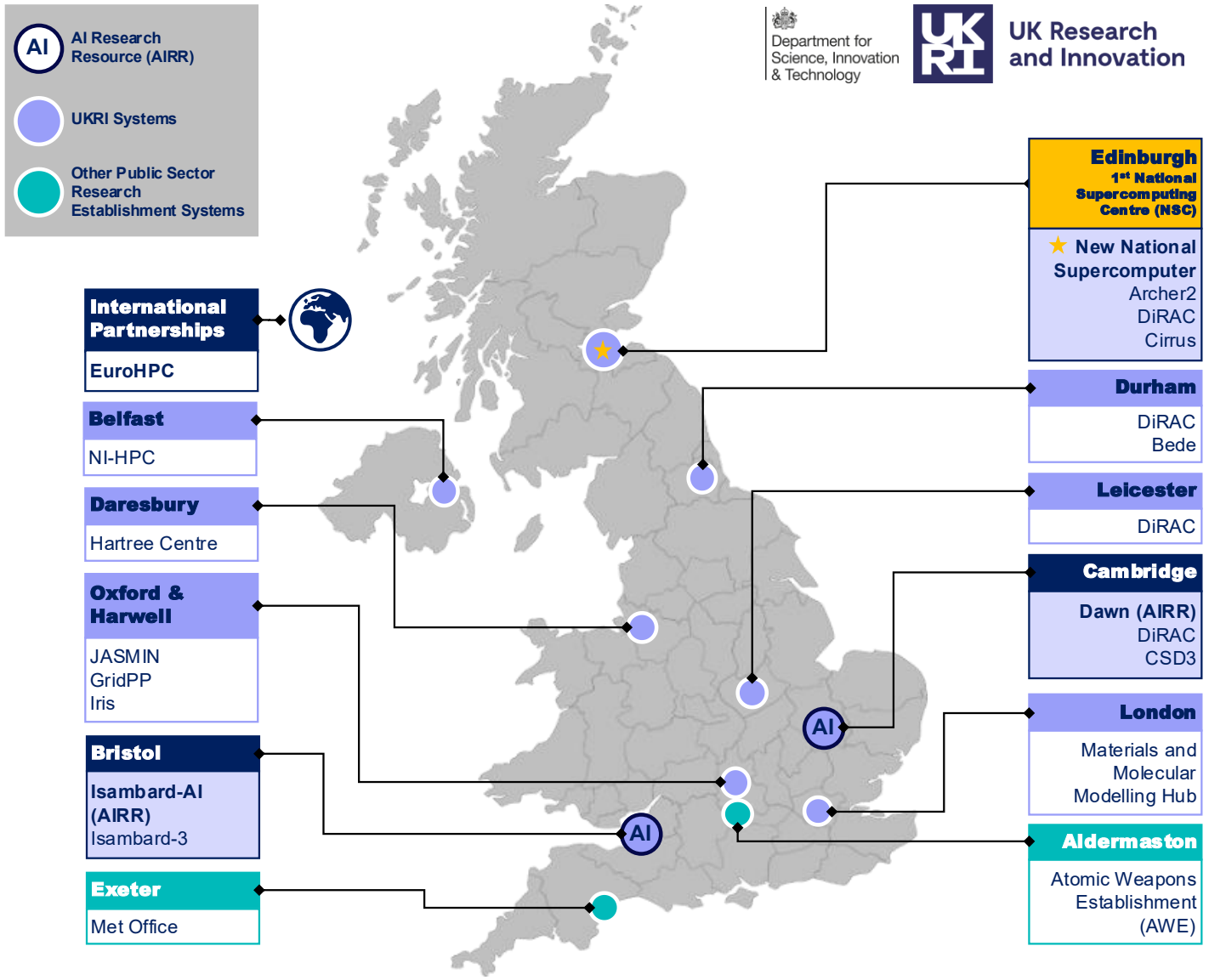
Tier-2 computers are fundamental for UK high performance computing as they provide a range of computing architectures which are driven by science needs not met by national or university-based facilities



The centres are:

1. **Isambard 3 (GW4)**
2. Kelvin-2 (NI-HPC)
3. JADE (Oxford)
4. Cirrus (EPCC)
5. Bede (N8 CIR)
6. CSD3 (Cambridge)
7. Baskerville (Birmingham)
8. Sulis (HPC Midlands+)

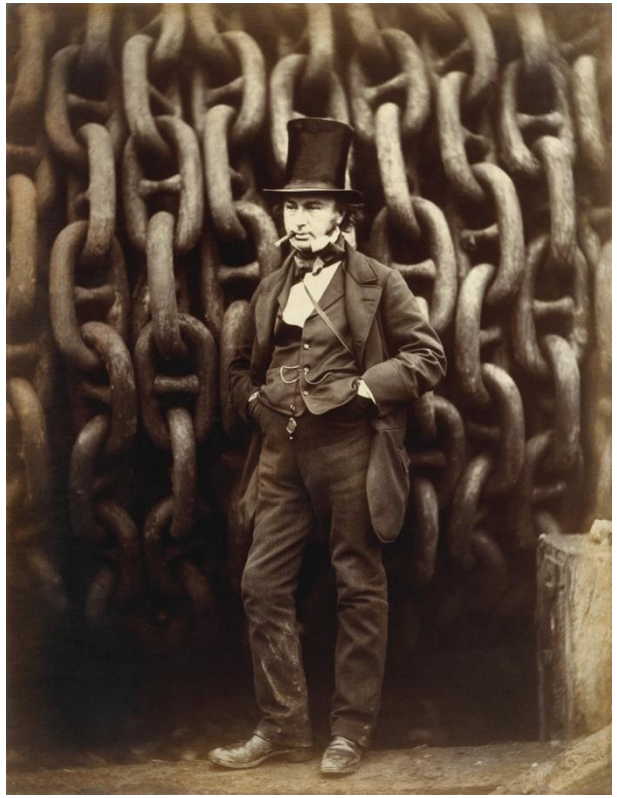
The UK e-infrastructure landscape



Tier-2 HPC Centres – Isambard 3

Isambard 3 is a £10 million, next-generation supercomputer that was developed as part of a collaboration between the **GW4** universities (Bath, Bristol, Cardiff, and Exeter) and is used for AI and high-performance computing (HPC) research.

Cardiff, through the GW4 alliance, has access to 10% of the compute resource available on Isambard 3 (384 NVIDIA Grace CPU Superchip nodes).



https://docs.isambard.ac.uk/

GW4

National Compute Resources (NCRs)

- Driven by the **UK Compute Roadmap** - a requirement for a compute ecosystem that is diverse, user-centred, coherent & serves the needs of the **UK's research and innovation communities** across a **wide range of disciplines and use cases**.
- As part of this national effort to develop an integrated public compute ecosystem, UKRI announced in October 2025 the plan to invest in **National Compute Resources (NCRs).** NCRs complement flagship investments in the **Next National Supercomputing Service and AI Research Resource** (AIRR).
- Provide **smaller scale services to ensure** the UK has the necessary diversity of skills & technologies – & the **expertise in specialised & novel solutions** – to enable cutting-edge research and innovation, increase resilience, & enable users to exploit new approaches.

https://engagementhub.ukri.org/ukri-infrastructure/phase-one-of-ukri-national-compute-resources/

Supercomputing Wales

- ARCCA is and has been part of different HPC collaborations
  - **HPC Wales**, a £44 million five-year project (2010–2015) to provide Wales with a world class HPC facility.
  - **Supercomputing Wales**, a £22 million five-year programme, built on success of the original HPC Wales project, 2018-2023
- ARCCA keeps close collaborations with Supercomputing Wales institutions (Aberystwyth, Bangor, Cardiff and Swansea Universities).

PRIFYSGOL ABERYSTWYTH UNIVERSITY

PRIFYSGOL BANGOR UNIVERSITY

CARDIFF UNIVERSITY
PRIFYSGOL CAERDYDD

Prifysgol Abertawe
Swansea University



arcca Supercomputing Wales overview



£22m programme

2016-2023 with initial £9.9m ERDF grant through Welsh Government

7,700

HPC users registered across a consortium of four universities in Wales





33,800 cores

cores in two upgraded HPC hubs delivering **2 Pflop**

2 Pflop

of compute power across both hubs, with access to >2 petabytes storage





15 RSEs

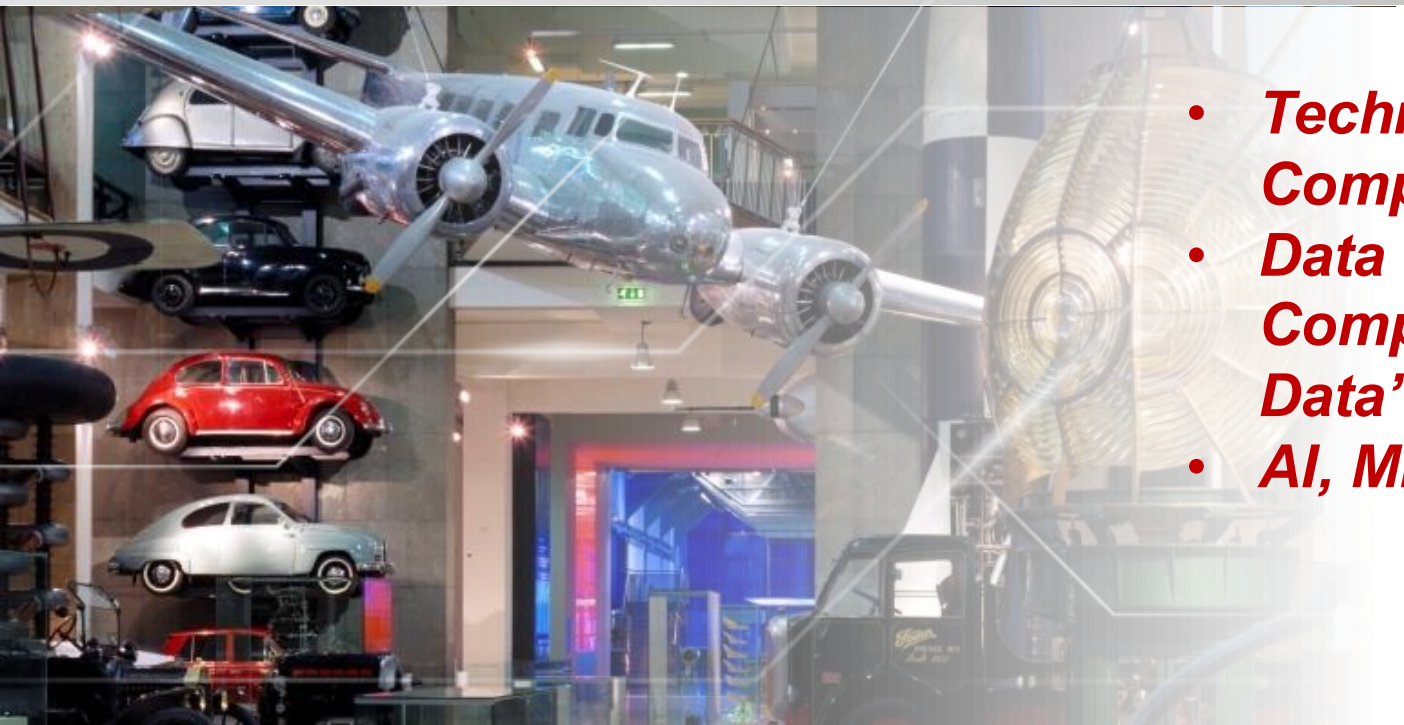
in four universities, working across multiple research domains

55 PhDs

funded through UKRI CDT in AI, Machine Learning & Advanced Computing



Why Advanced Research Computing?



- *Technology, Parallel Computing*
- *Data Intensive Computing – "Big Data"*
- *AI, ML & Deep Learning*

What is Advanced Research Computing?

Broadly, we define Advanced Research Computing to be the use of computing and data resources for research, which are beyond the capabilities of the average desktop or laptop computer.

Use of leading-edge IT resources & tools to pursue research;

1. ***Computer simulation and modelling***
   - insight into problems where experiment is not feasible
   - the modelling of whole systems as well as system components
   - problems with a no. of different length (and perhaps time) scales
2. ***Manipulating and storing large amounts of data***
   - processing research data and analysis at high speeds
   - the storage of large amounts of data for future analysis
   - **Data analytics, machine and deep learning capability**
3. ***Other methods to solve complex research problems***
   - a network capability linked to other computers to allow access & analysis [*Cloud Computing*]
   - the visualisation of data in formats that explain outcomes

What can we do with Faster Computers?



JUPITER (Joint Undertaking Pioneer for Innovative and Transformative Exascale Research) is Europe's first **exascale supercomputer**, located at the Forschungszentrum Jülich campus in Germany. Inaugurated in September 2025, it is the most powerful computer in Europe and ranks 4th on the June 2025 TOP500 list of the world's fastest supercomputers.

- Solve problems faster
  - Reduce turn-around time of big jobs
  - Increase responsiveness of interactive applications
- Get “better” solutions in same amount of time
  - Increase resolution of models
  - Make model more sophisticated

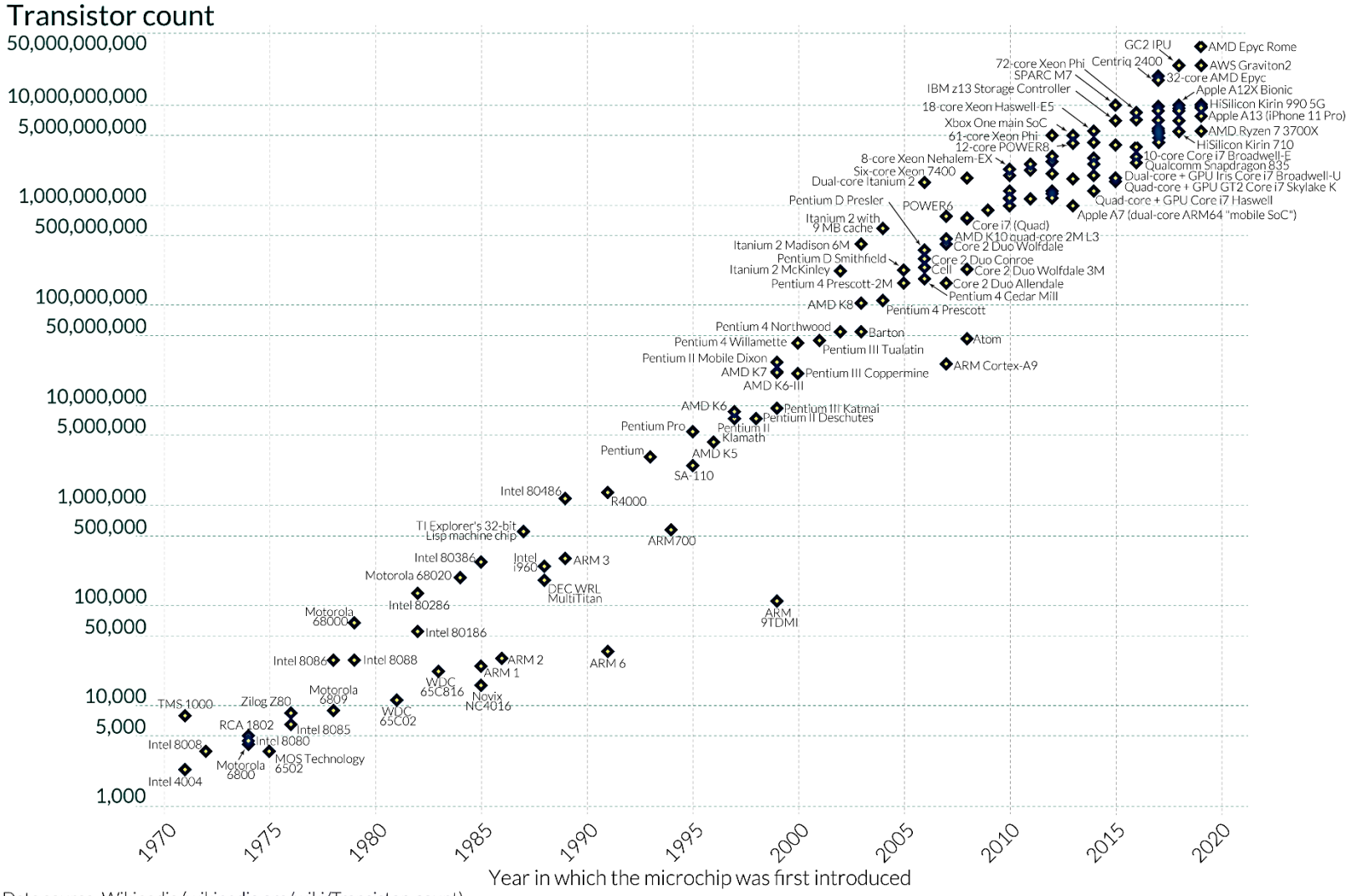
….. Memory and Storage requirements ?

"It costs £500,000 to do each physical test of a car crash, and it's not repeatable. It costs £12 to run a virtual simulation of a car crash, and it’s fully repeatable, so it can be used to optimise the design of a vehicle." Andy Searle, Head of Computer Aided Engineering, Jaguar Land Rover.

Moore's Law: The number of transistors on microchips has doubled every two years

Our World in Data

Moore's law describes the empirical regularity that the number of transistors on integrated circuits doubles approximately every two years.
This advancement is important for other aspects of technological progress in computing – such as processing speed or the price of computers.



Data source: Wikipedia (wikipedia.org/wiki/Transistor_count)
OurWorldinData.org – Research and data to make progress against the world's largest problems.

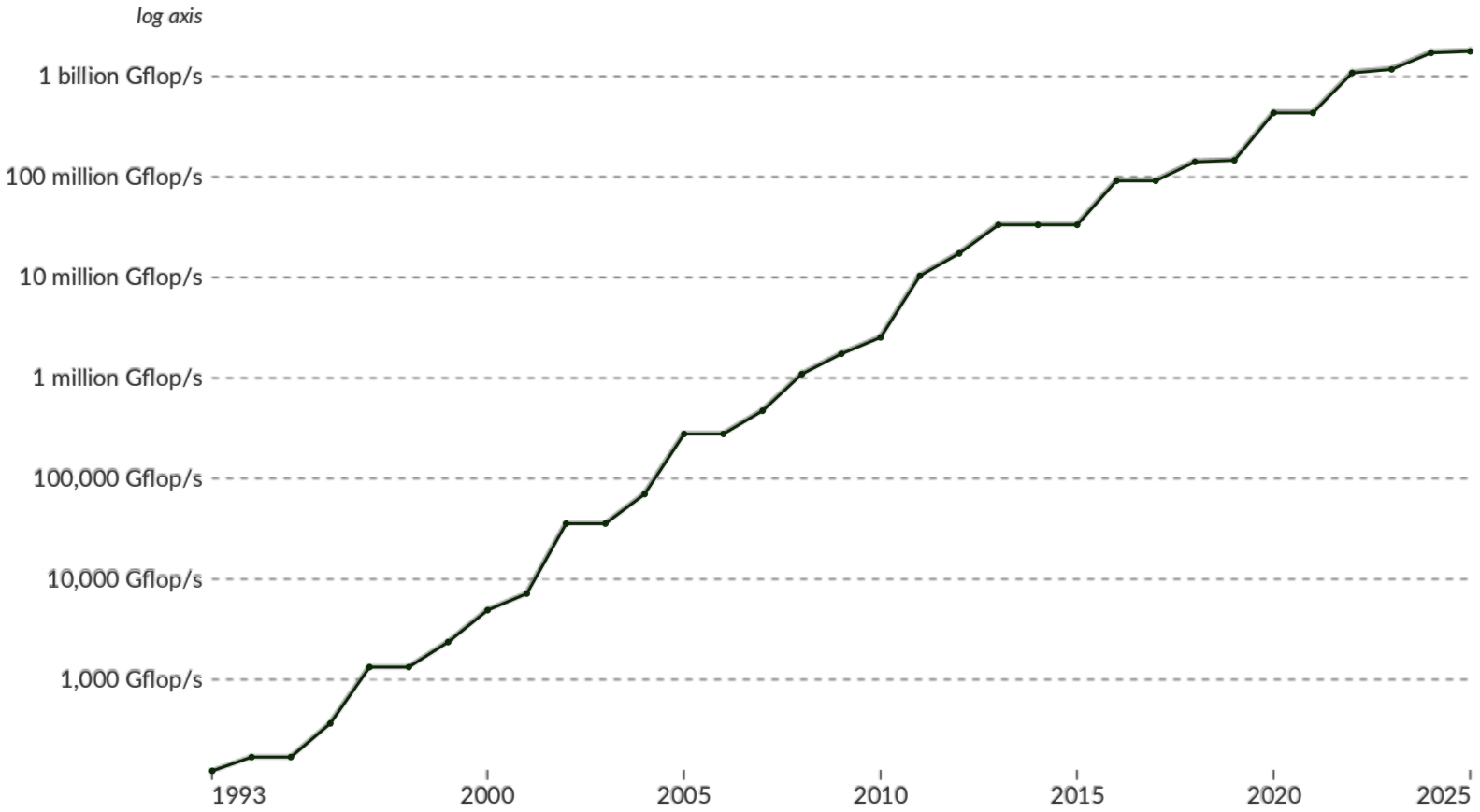
Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

What can we do with Faster Computers?

Computational capacity of the fastest supercomputers

Our World in Data

The number of floating-point operations[1] carried out per second by the fastest supercomputer in any given year. This is expressed in gigaflops, equivalent to 10^9 floating-point operations per second.

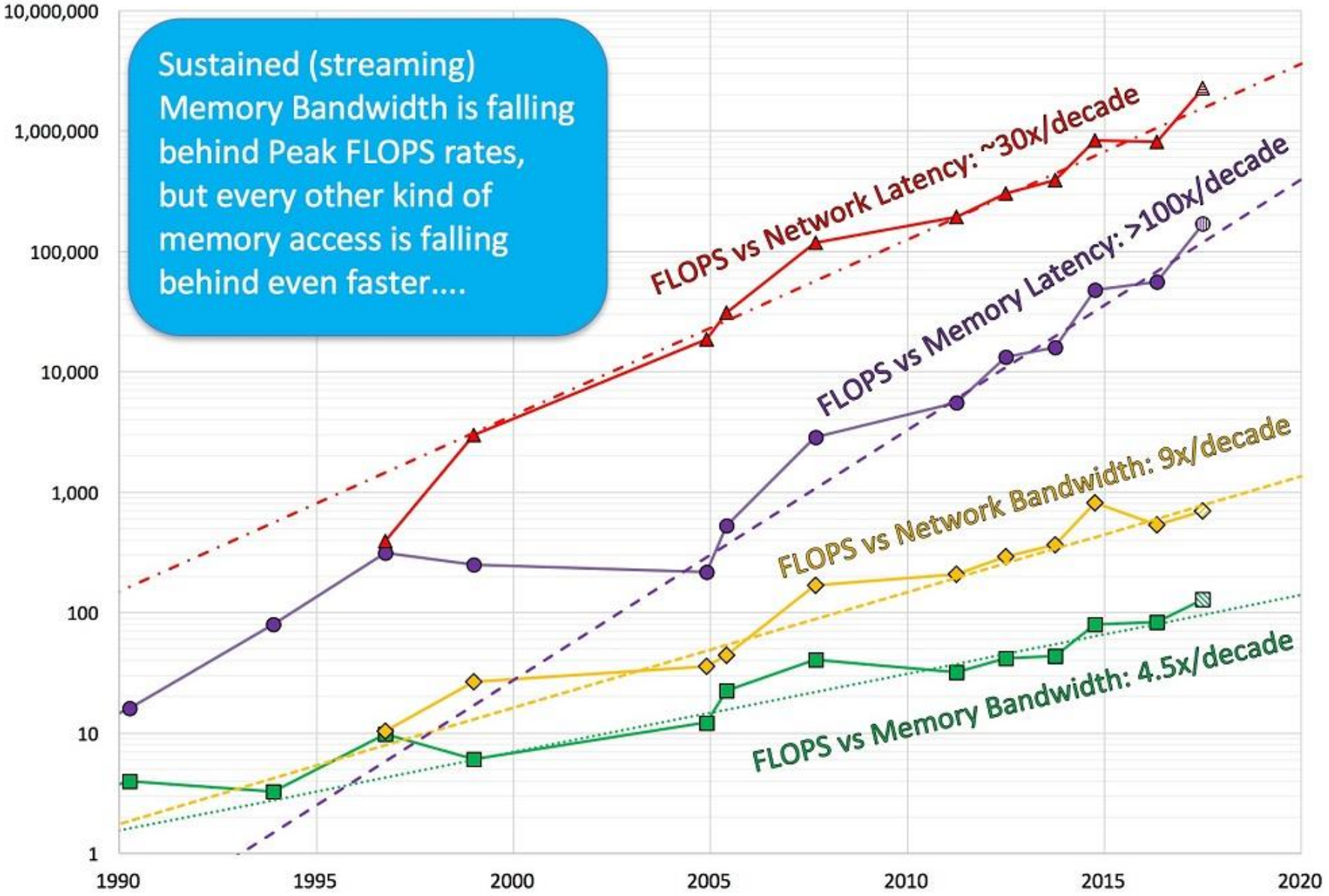


Data source: Dongarra et al. (2025)

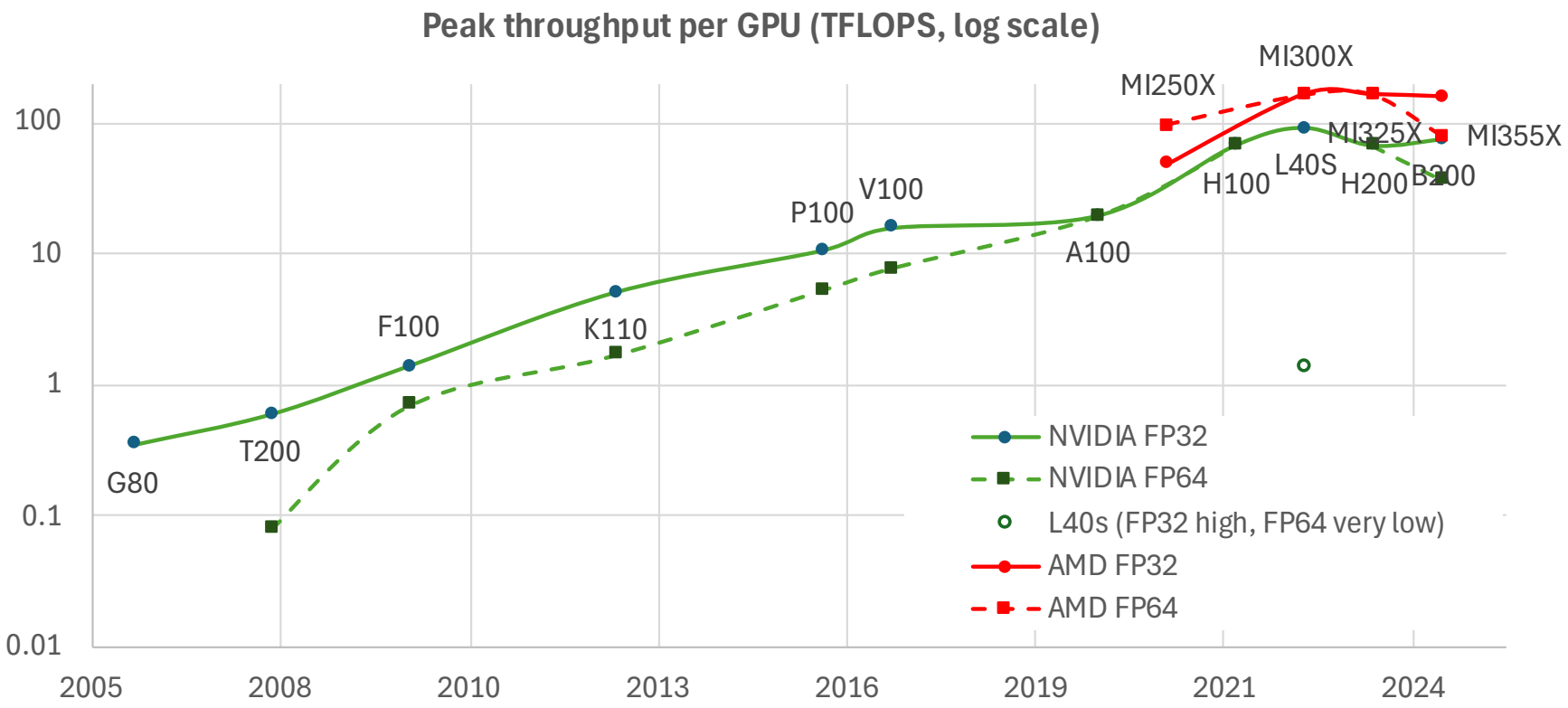
OurWorldinData.org/technological-change | CC BY

1. **Floating-point operation** A floating-point operation (FLOP) is a type of computer operation. One FLOP represents a single arithmetic operation involving floating-point numbers, such as addition, subtraction, multiplication, or division.

Memory Access and Peak Mflop Rates



Single GPU throughput



Sources: NVIDIA and AMD published specifications. NVIDIA 2006-2020 values read approximately from the original throughput chart published by W. J. Dally, S. W. Keckler and D. B. Kirk, "Evolution of the Graphics Processing Unit (GPU)," in IEEE Micro, vol. 41, no. 6, pp. 42-51, 1 Nov.-Dec. 2021, doi: 10.1109/MM.2021.3113475.
H200 has the same compute as H100 with 4.8 TB/s memory.
FP64 shown is the Tensor-Core (NVIDIA) or Matrix-Core (AMD) rate.
L40S FP64 TFLOPS are estimated at 1/64th of single precision as the device does not feature native double-precision (FP64) tensor or CUDA cores.

The Scientific Paradigm has grown

The Evolution of Research Computing

- **Reshaping the User Experience** – Interface Choices;
- **HPC Storage** – data-driven requirements;
- **AI driven requirements:** Accelerator technologies;



Interface Choices

- The **OnDemand service** was adopted by ARCCA a few years back as a response to an increased number of requests to **support interactive software development tools** (e.g. JupyterLab, Rstudio).
- A combination of **new documentation** for the service and **personalised support sessions** has helped users migrate existing workflows to the supercomputer.
- The underlying queuing system that OnDemand currently deploys on the supercomputer leave **areas of opportunity to improve the user experience**.











Data Driven Requirements

1. **Systems are generating higher volumes of data. This needs higher performance, to understand, predict and manufacture with higher fidelity**
   - **Accelerators** are becoming essential; Higher storage performance and scalability required to store and serve data; More advanced networking capability is needed to move data
2. **Systems are becoming increasingly complex, and users are looking to simplify operations**
   - Workloads as a service: Rapid changes in technology making customer decisions difficult
3. **Data is generated at multiple places, in a distributed fashion**
   - Desire to make decisions closer to data source or the decision point - **Higher bandwidth connection between data sources is needed**
   - Management of data & infrastructure becoming more critical. Key driver in requirements analysis
4. **Automation and AI are driving significant change in IT architecture**
   - This will continue to be a major technology driver

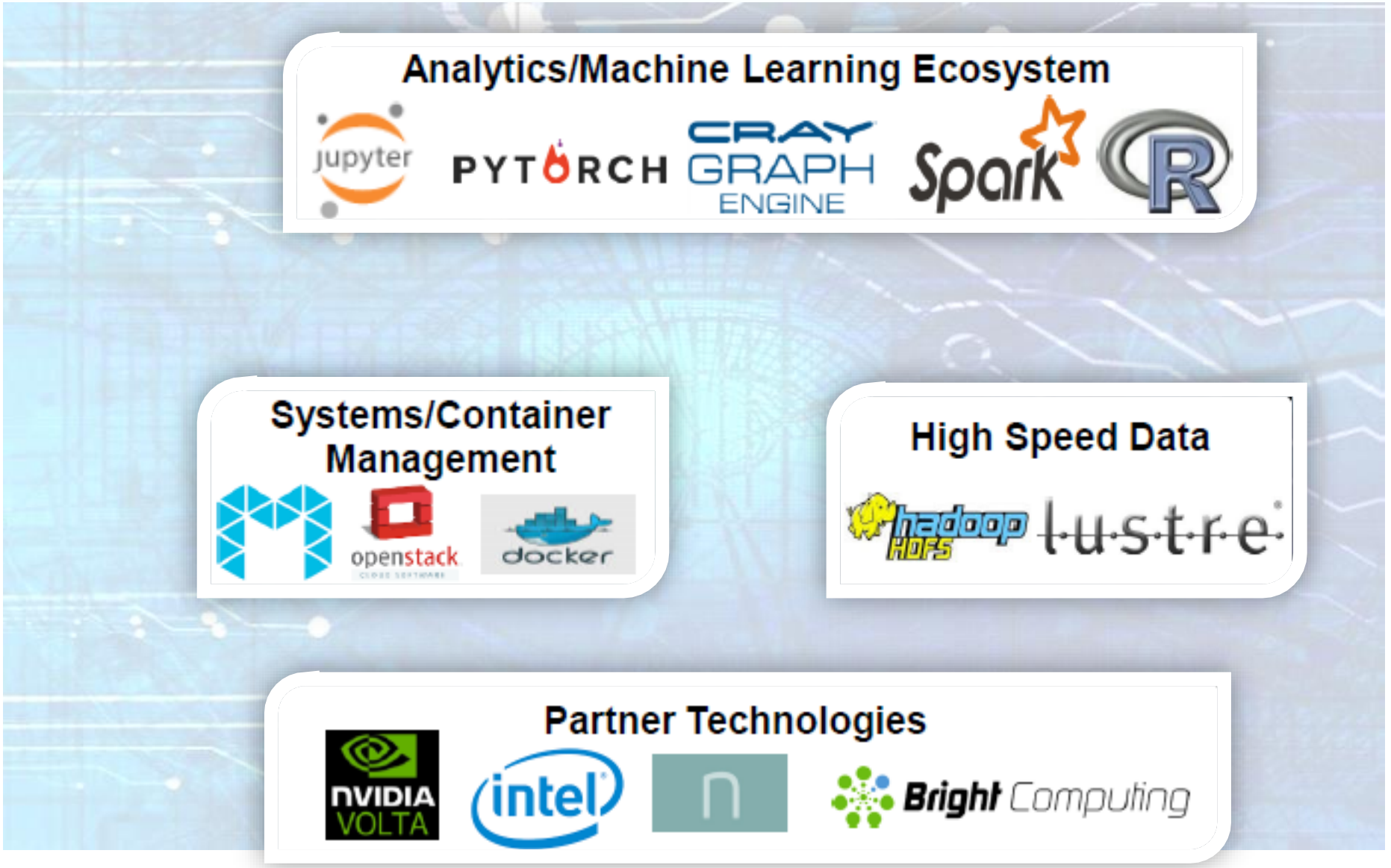
AI driven requirements – Supercomputing Wales

Increased use of GPU accelerated workflows by Machine and Deep Learning. This has led to new developments and opportunities:

- Development of in-house expertise around **accelerator technologies**.
- Incorporation of **improved system performance monitoring (NVIDIA DCGM) and optimization tools (nvprof, nsight)**
- Expansion of available documentation for the use of GPUs on the supercomputer and associated work environments.
- Certified **NVIDIA DLI Ambassadors Training** and new Courses
- Increasing **Understanding of user expectations & adaptation of user interfaces**.

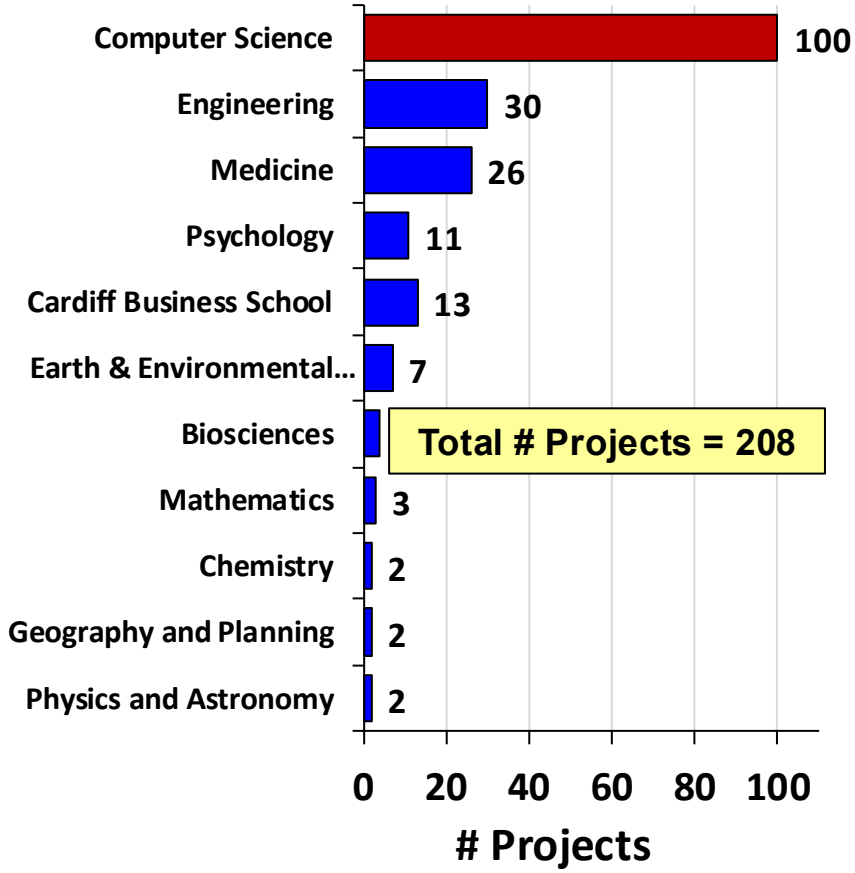
	Projects (Total)	Projects (COMSC	Usage (core hrs)
Aberystwyth	55	42	2,034,230
Bangor	41	15	5,750,540
Cardiff	208	100	9,913,968
Swansea	87	33	12,342,917
Total	391	190	30,041,655

arcca AI Environment

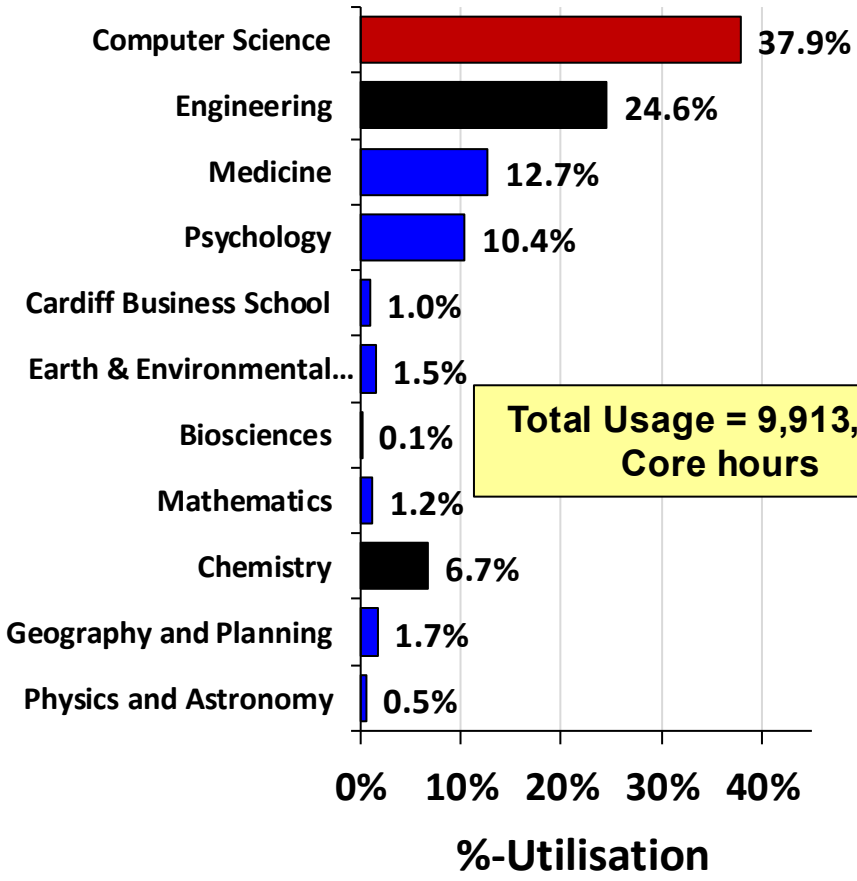


arcca SCW AI Project Profile and Usage - Cardiff

AI Project Profile



AI Utilisation Profile



Total # Projects = 807

Total Usage = 748,985,623 Core hours

What is Parallel Computing ?

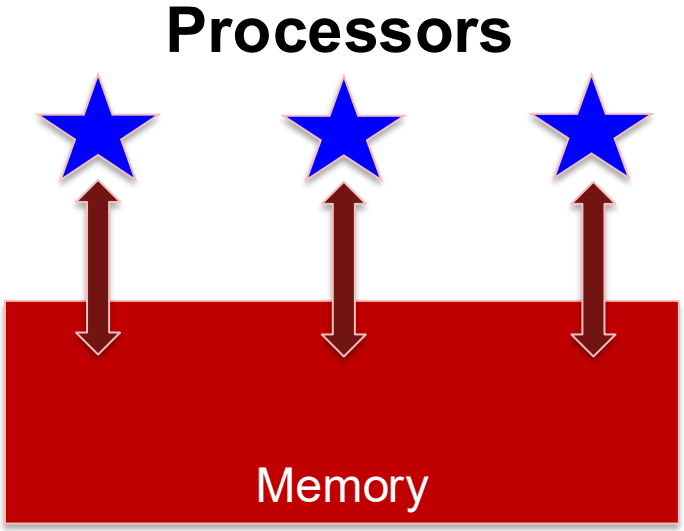


The Software Challenge and a word of caution ....

What is Parallel Computing?

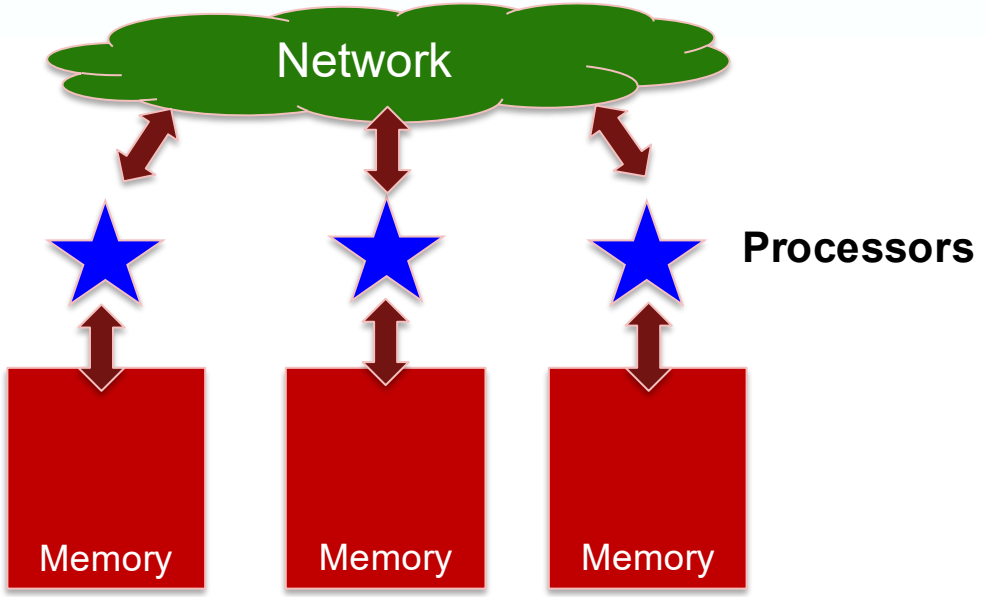
- A strategy for performing large, complex tasks faster.
- Clock speeds no longer increasing exponentially
  - Problems caused by higher speeds; Excessive power consumption, Heat dissipation, Current leakage
- A large task can either be performed serially, one step following another, or can be decomposed into smaller tasks to be performed simultaneously, i.e., in parallel.
  - One might hope that given P processors you can solve your problem P times quicker …
- Attempt to speed solution of a particular task by
  1. Dividing task into sub-tasks
  2. Executing sub-tasks simultaneously on multiple processors

Shared and Distributed Memory



Shared memory

Technologies such as OpenMP allow shared memory to be easily programmed



Distributed memory

Message Passing Interface (MPI):

- Portable and available for C and Fortran
- Decomposing data: Processors working on same problem (but different views);
- Works within processors on node and across networks.
- Limited only by Amdahl's Law

Amdahl's Law

Amdahl's Law places a strict limit on the speedup that can be realized by using multiple processors.

- Effect of multiple processors on run time

$$t_n = (f_p / N + f_s) \times t_1$$

- Where
  - f_s = serial fraction of code; f_p = parallel fraction of code
  - N = number of processors; t_1 = time to run on one processor

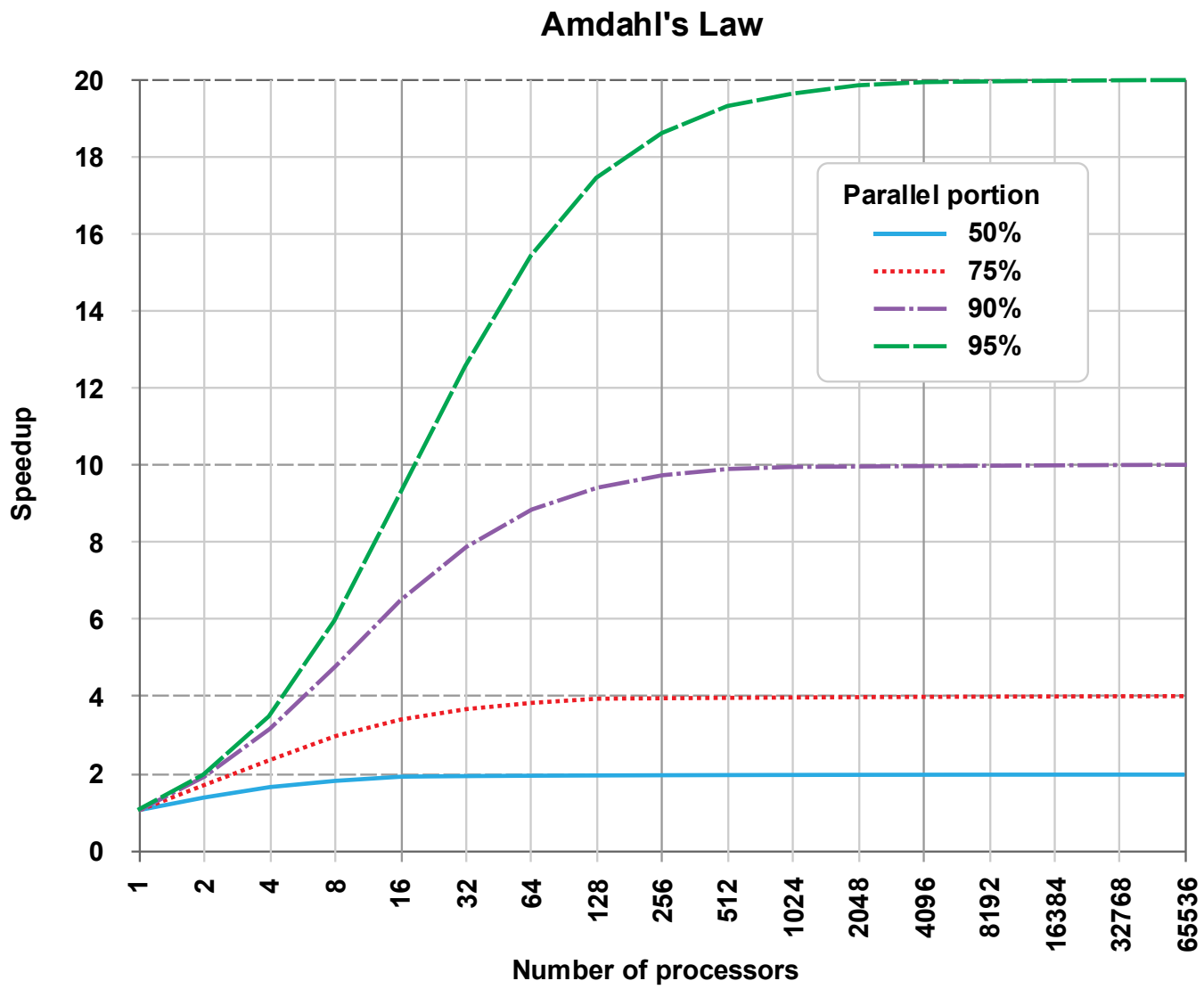
Speed up formula:

$$S = 1 / (f_p / N + f_s)$$

f_s = serial fraction of code;
f_p = parallel fraction of code;
N = number of processors

- **Case:**
  1. $f_s = 0$, $f_p = 1$, then S = N
  2. N → infinity: $S = 1/f_s$; if 10% of the code is sequential, you will never speed up by more than 10, no matter the number of processors.

Amdahl's Law



Source: https://en.wikipedia.org/wiki/Amdahl%27s_law

Classes of Parallelism

Successful attempts require both

1. Understanding of where parallelism can be effective
2. Knowledge of how to design and implement good solutions

Data Parallelism: It focuses on distributing the data across different nodes, which operate on the data in parallel.

Task (Functional) Parallelism: is distinguished by running many different tasks at the same time on the same data. These are often called "***Embarrassingly Parallel***".

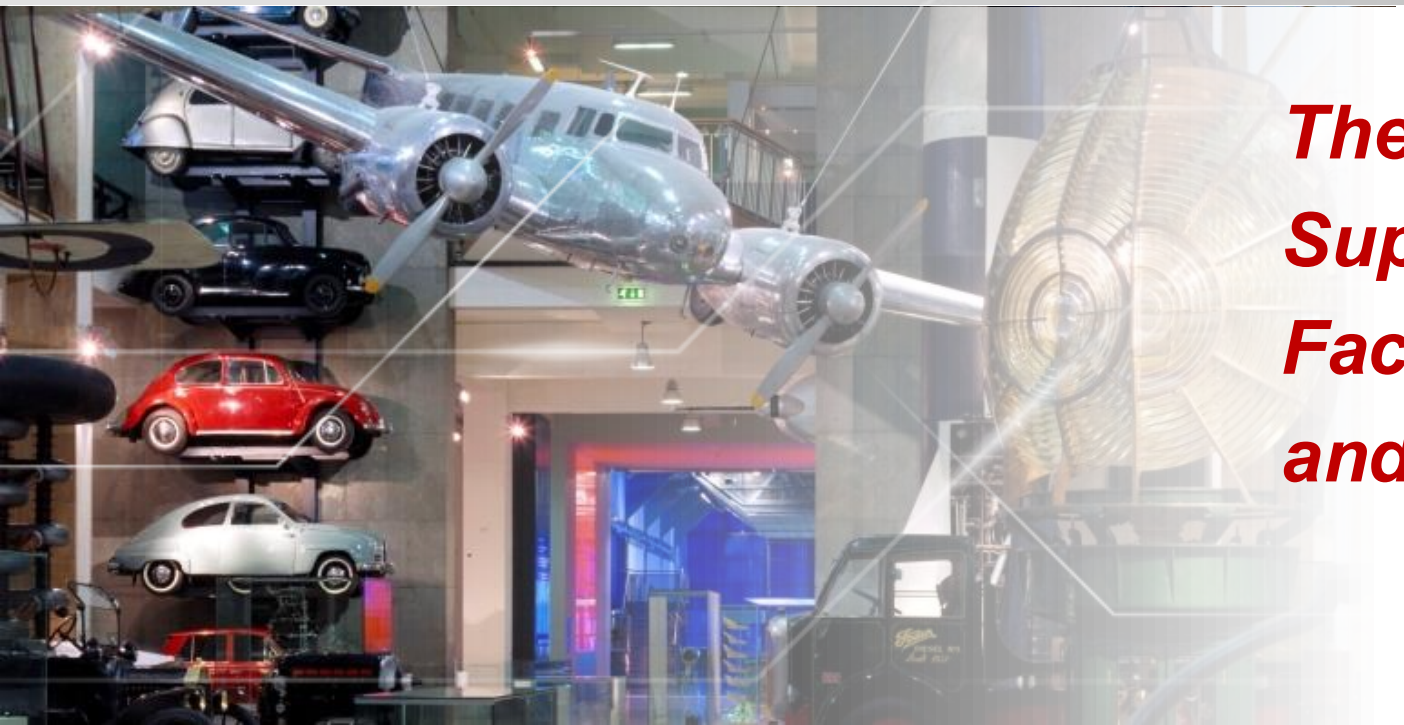
Pipeline Parallelism: each stage works on a part of a solution. The output of one stage is the input of the next; this works best when each stage takes the same amount of time to complete.

Is it really worth it to go Parallel?

- **Writing effective parallel applications is difficult !!**
  - **Load balance** is important
  - **Communication** can limit parallel efficiency
  - **Serial time** can dominate
- **Is it worth your time to rewrite your application?**
  - Do the CPU requirements justify parallelization? Is your problem really `large'?
  - Is there a library that does what you need (e.g., parallel FFT)
  - Will the code be used more than once?

- All parallel programs contain:
  - parallel sections (we hope!) and serial sections (unfortunately)
- Serial sections limit the parallel effectiveness
- **Amdahl's Law** states this formally

Supercomputing at Cardiff University



The Cardiff Supercomputing Facility: Projects, and Highlights

Advanced Research Computing @ Cardiff

Established in 2007 to "transform the University's approach to advanced research computing" and to



- **Bring together the advanced research computing community**
- **Introduce and encourage the use of new techniques and technologies**
- **Encourage and enable new user communities and new applications**
- **Encourage new and interdisciplinary research using advanced research computing techniques**

To build and sustain a position which takes the university to the forefront of leading research universities in the UK and internationally in this field

Provide a responsive support infrastructure that addresses the key requirement of enabling the research of others.

Expanded Capability



Impressive expansion since the beginning of **Supercomputing Wales (SCW)** and the installation of **"Hawk",** Falcon's predecessor in Q3 2018, including:

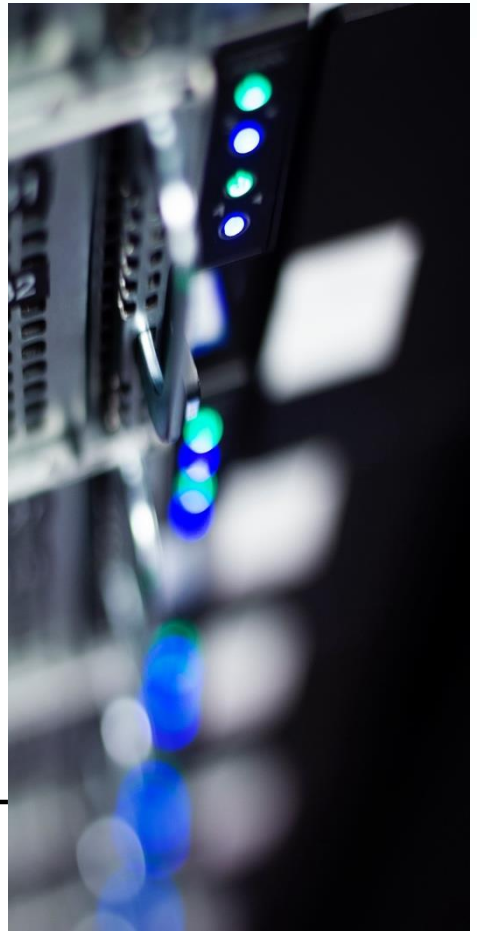
- Seven-fold user growth – from 700+ in March 2019 to > 4,700 users
- Hawk's size and complexity increased to support dedicated researcher funded expansions (from 8,000 (Intel) processing cores in 2018 to over 25,000 (Intel & AMD) prior to decommissioning +GPUs for AI/Deep Learning.
- Enhanced Productivity, since 2018 leveraging > £200 Million (M) competitively sourced income for research projects (FEC £223M) dependent on access to these facilities.
- We expect these trends to continue with the release in Q3 2026 of our new HPC system **Falcon**.

Services Evolution

Historically focused on traditional supercomputing provision, application support, training and engagement.

Supercomputing provision has become ***much more complex*** over last 7 years:

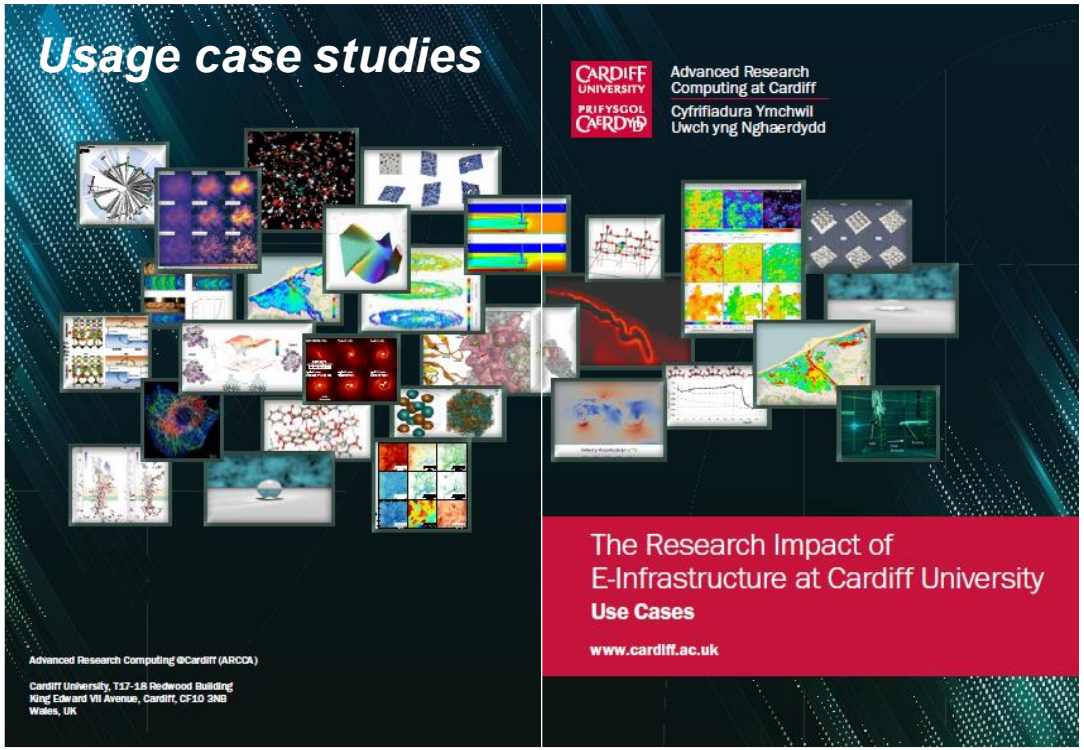
- **Core Infrastructure** using proven cluster management middleware comprising management, login nodes, compute & high performance parallel file systems.
- **Researcher Funded Partitions** Integration – Wales Gene Park, LIGO, Chemistry, Computer Science, Dementia Research Institute, Data Innovation Institute etc. with dedicated partitions. Demanded expansion of the Infiniband fabric (x2)
- **Gluster Storage** – data staging area (interim solution – semi-permanent service?)
- **Web Interface** access points (simplifying user access)
- **Virtual Infrastructure** Integration
- **Transitioning research computing workloads to the Cloud**
- **Evolving AI requirements and Integration**





SUNBIRD

Research Impact and Outcomes

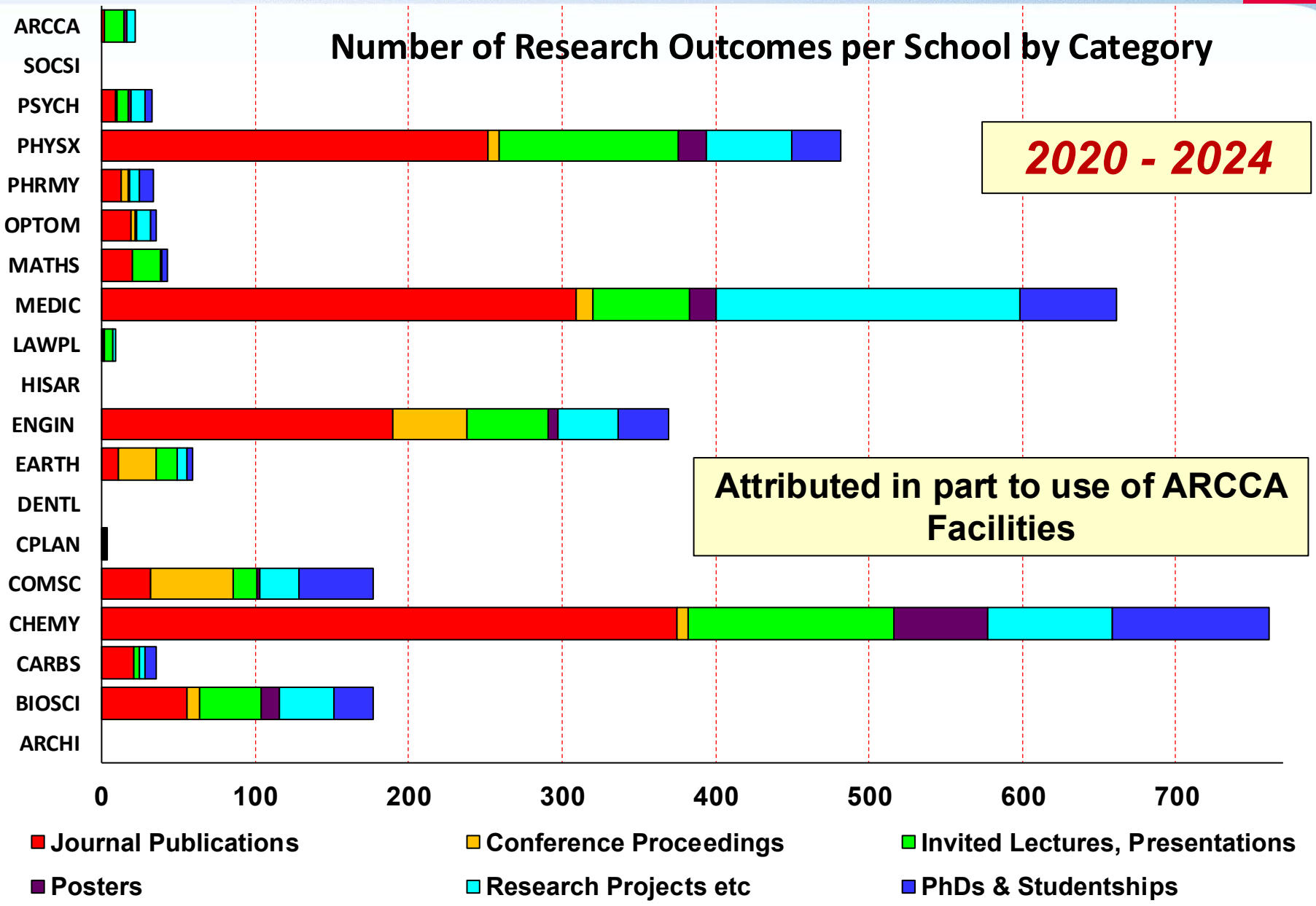


80 Use Cases from the following Schools

- Biosciences
- Business School
- Chemistry
- Computer Science & Informatics
- Earth and Ocean Sciences
- English, Communication & Philosophy
- Engineering
- Medicine
- Optometry & Vision Sciences
- Pharmacy
- Physics and Astronomy
- Psychology
- Social Sciences

Analysis: Data points to Cardiff University leveraging since 2018 > **£200.0** Million (M) of competitively-sourced income for research projects (FEC **£222.7M**) dependent on access to supercomputing facilities

Research Outcomes per School by Category



Cardiff Schools Utilisation of Hawk

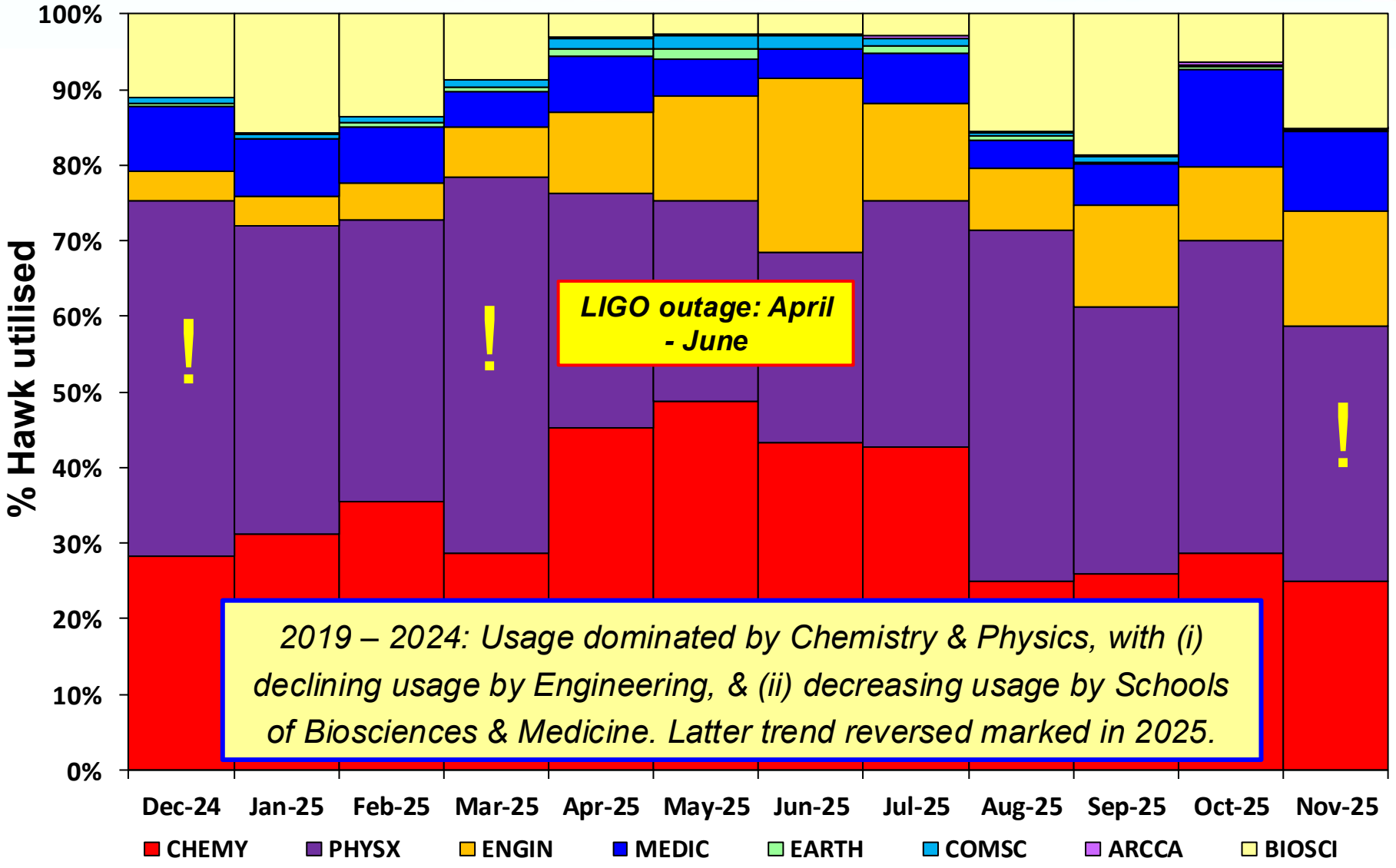


Figure - Percentage utilisation of Hawk by School.

The Falcon Supercomputer



The ARCCA Supercomputer Service

Compute (CPU)

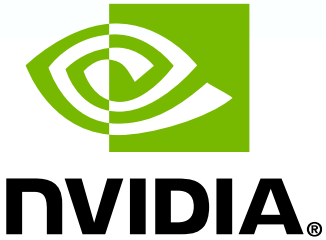
1. **~9,000 AMD cores (EPYC 9654 "Genoa")**
   - 39 x 192 cores, RAM 755Gi
   - 8 x 192 cores, RAM 1.5 Ti ("highmem")
2. **~ 2,000 AMD cores (EPYC Rome 7502 "Rome")**
   - 32 x 64 cores, RAM 250Gi



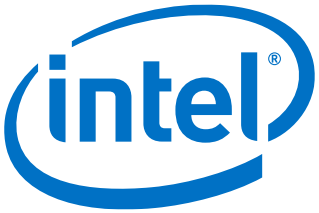


AMD EPYC

Accelerators



- NVIDIA GPUs
  - **1 x 4-way H100 (80 GB) node** (Intel Emerald Rapids 6530 @ 4 GHz)
  - **1 x 4-way H200 (140GB) node** (Intel Emerald Rapids 6530 @ 4 GHz)
  - **2 x 8-way L40S (48 GB) nodes** (Intel Saphire Rapids 6430 @ 3.4 GHz)
  - **15 x 2-way V100 (16 GB) nodes** (Intel Cascade Lake Gold 6248 CPU @ 2.40GHz)



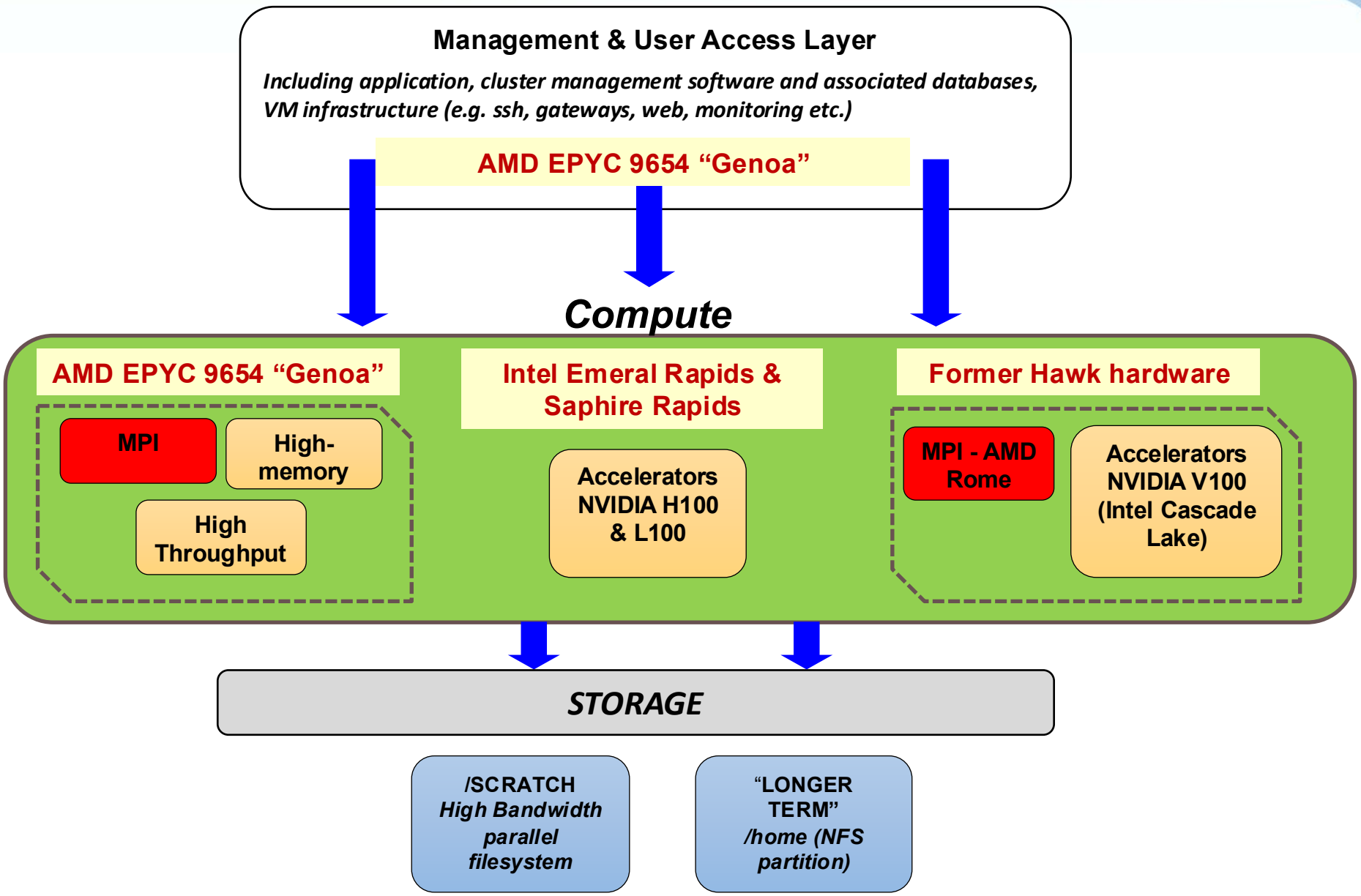
The Falcon Supercomputer - Storage

1. Storage has proven historically to be a challenging feature of the ARCCA Clusters, driven by 2 factors
   - Allowing users to keep extensive data sets on the /scratch file system given challenges faced by that community in finding alternative, fit-for-purpose solution elsewhere, and
   - The inability of the current RDS system to house anything beyond modest data sets, both in terms of limited capacity and excessive cost.



2. The current **Falcon storage provision** is little better than that on Hawk, its predecessor. Specifically
   - 1.2 PB, Scratch
   - 250 TB, Home
   - 250 TB, Applications
3. The pressure on the /scratch file system will be eased with the arrival of the new RDS system later in the calendar year

The Falcon Supercomputer



A "Pluggable" Infrastructure

Cardiff University researchers can purchase additional hardware (compute and storage) and connect it to Falcon so that it is available to support their work and the work of their colleagues

Established Benefits:

1. Systems **procured, then "fed and watered", by ARCCA Staff**, as part of the Falcon service – avoids separate procurement process (through Directly incurred (DI) provision)
2. **Dedicated resource** – avoids queuing regime of Falcon providing instant turn around – ideal for code/workflow development.



Examples:

Physics. Support for 1060 cores (Intel) in two partitions

Chemistry: Support for 1024 cores (AMD) across two partitions.

Cardiff University's Supercomputer Infrastructure



Data Centre Provisoning - From Redwood to Church Village

Cardiff's Supercomputer Infrastructure



Initially hosted in Cardiff University's Redwood Pharmacy building



High Density Servers

Data Centre Update

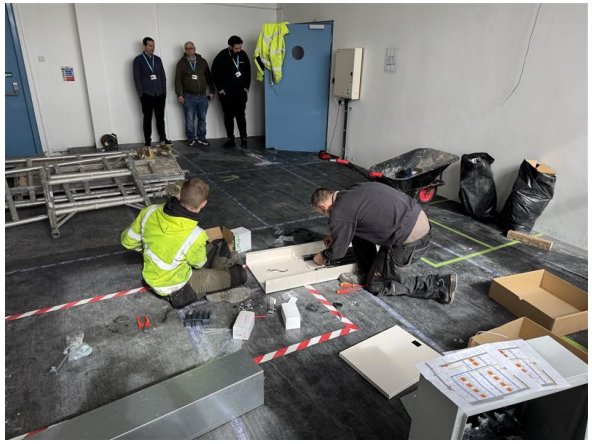
It has been migrated to Church Village (north of Cardiff).









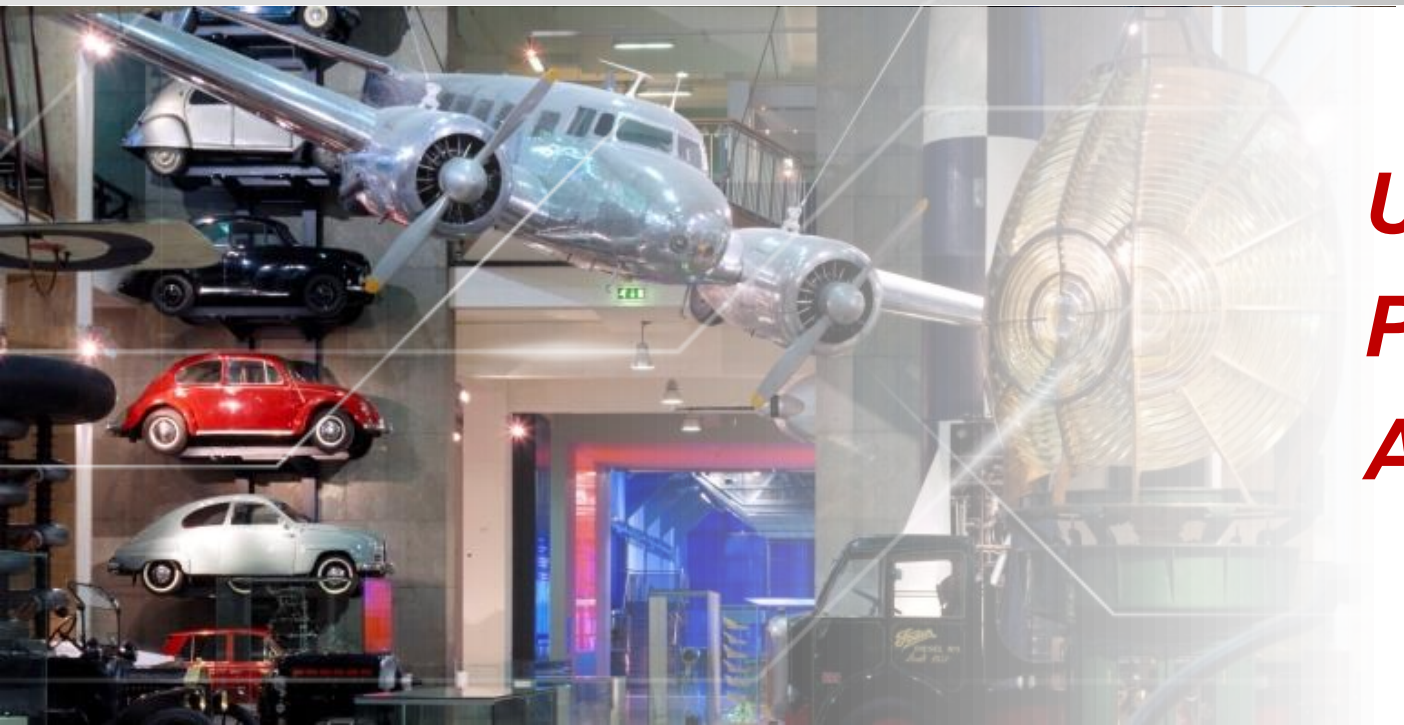


Cardiff's Hosting Capability

Cardiff now has access to a **new server room** at **CoLocation facility** for its immediate research computing needs (**circa 20 racks, 0.5MW**)

- We have **4 Direct Liquid Cooled (DLC) racks – expandable to 12**.
- There is **scope for expansion of services** in both the short and long term
  - **2MW – 3-6 months, dependent on method of delivery, circa £2m**
  - **10-20MW – 18 months, cost TBC**
- **Two options** being looked at for the **UKRI system**
  - Container in a secured area outside the main building (6 months)
  - Build something new (9 months)
- **Likely that the operating cost ceiling (£2m) is limiting factor for UKRI**

Accessing Falcon and getting started



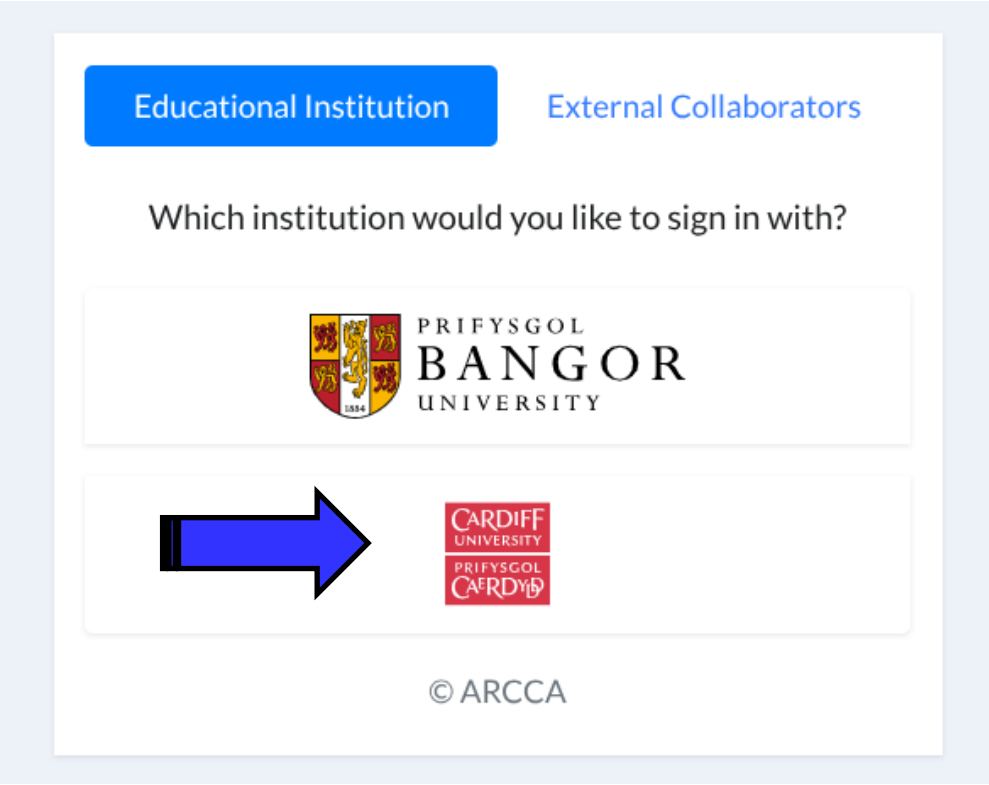
User and Project Accounts

Accessing Falcon

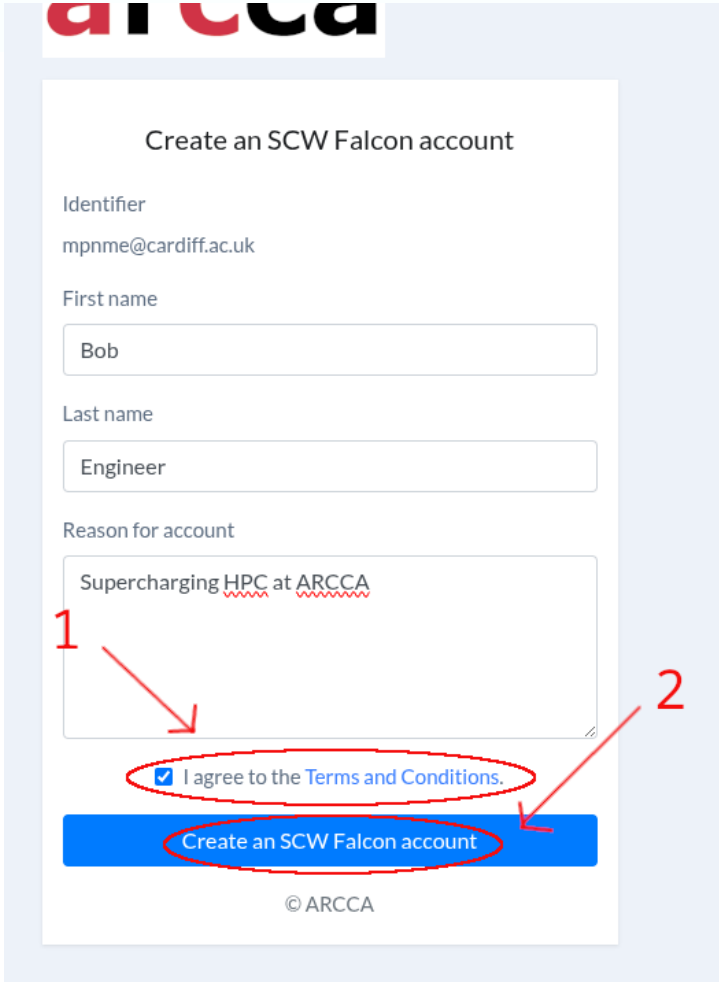
- A far from straightforward process given the transition from a **user driven approach** to address the requirements for a **Principal Investigator (PI), project driven regime** (imposed by University & UKRI funding)
- Request a **new user account** – this gives you access to the system. **Required by both PIs AND Users**
- **Request** (PIs) or **join** (non-PIs) a **project account** – this gives you the ability to run jobs on the system.
- **Read full instructions** in our documentation website to avoid unnecessary delays:

https://wiki.arcca.cf.ac.uk/index.php/The_Falcon_Supercomputer

New User Account I

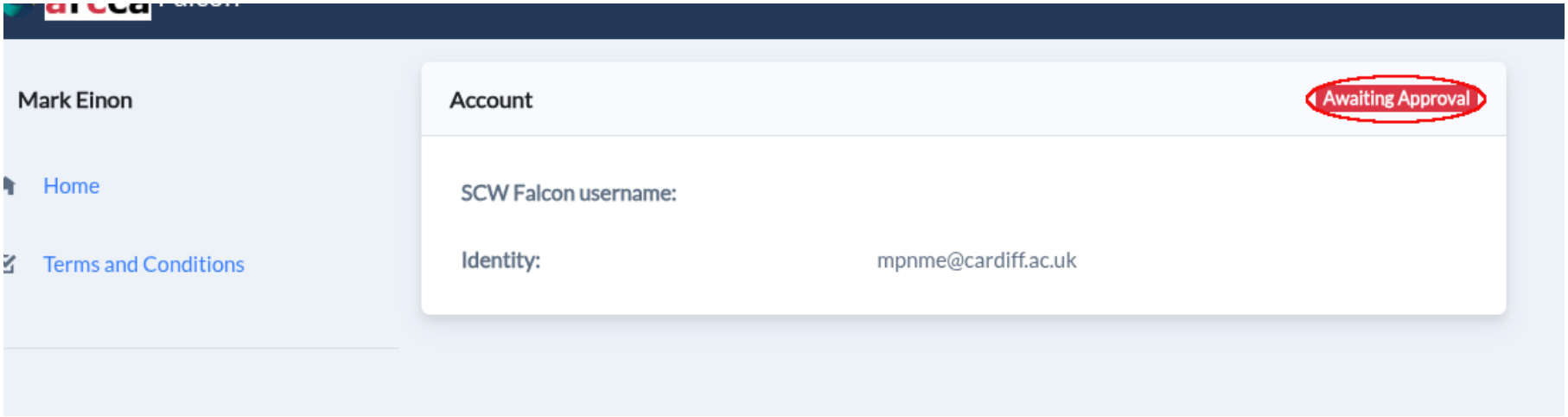


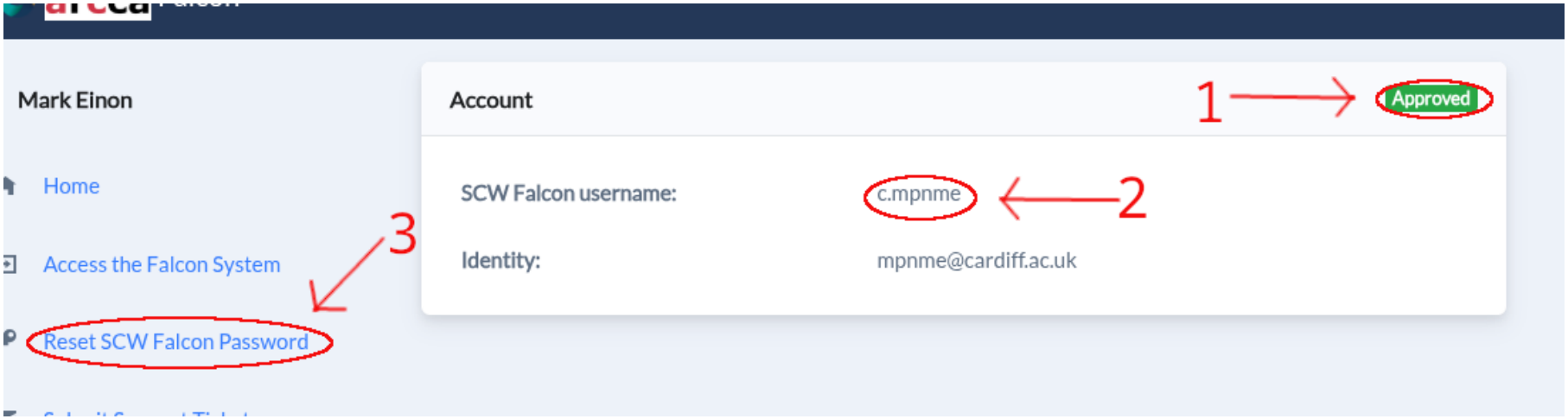
To register for an account visit COGS https://cogs.cf.ac.uk



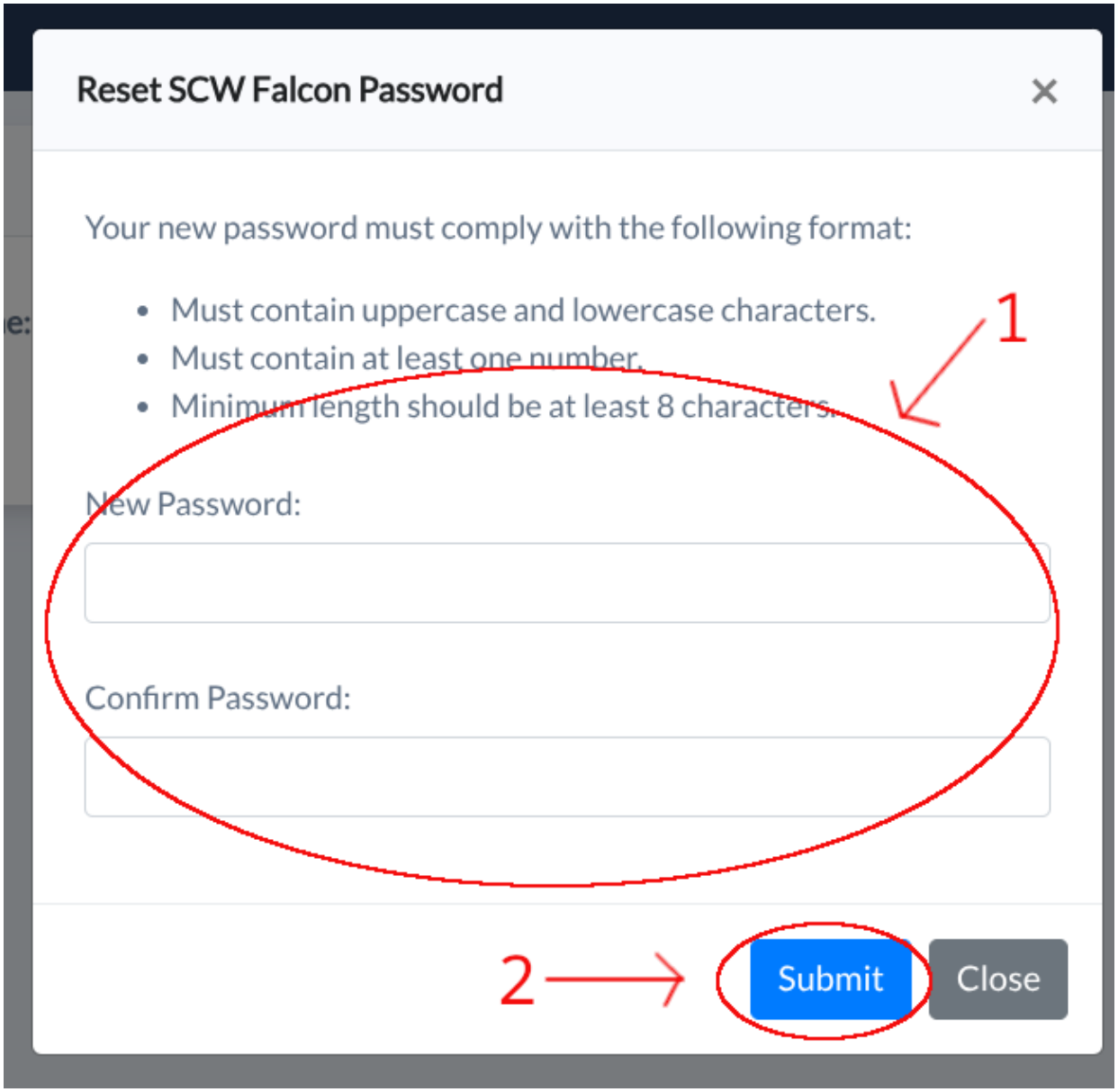
Read the **Terms and Conditions**

arcca New User Account II



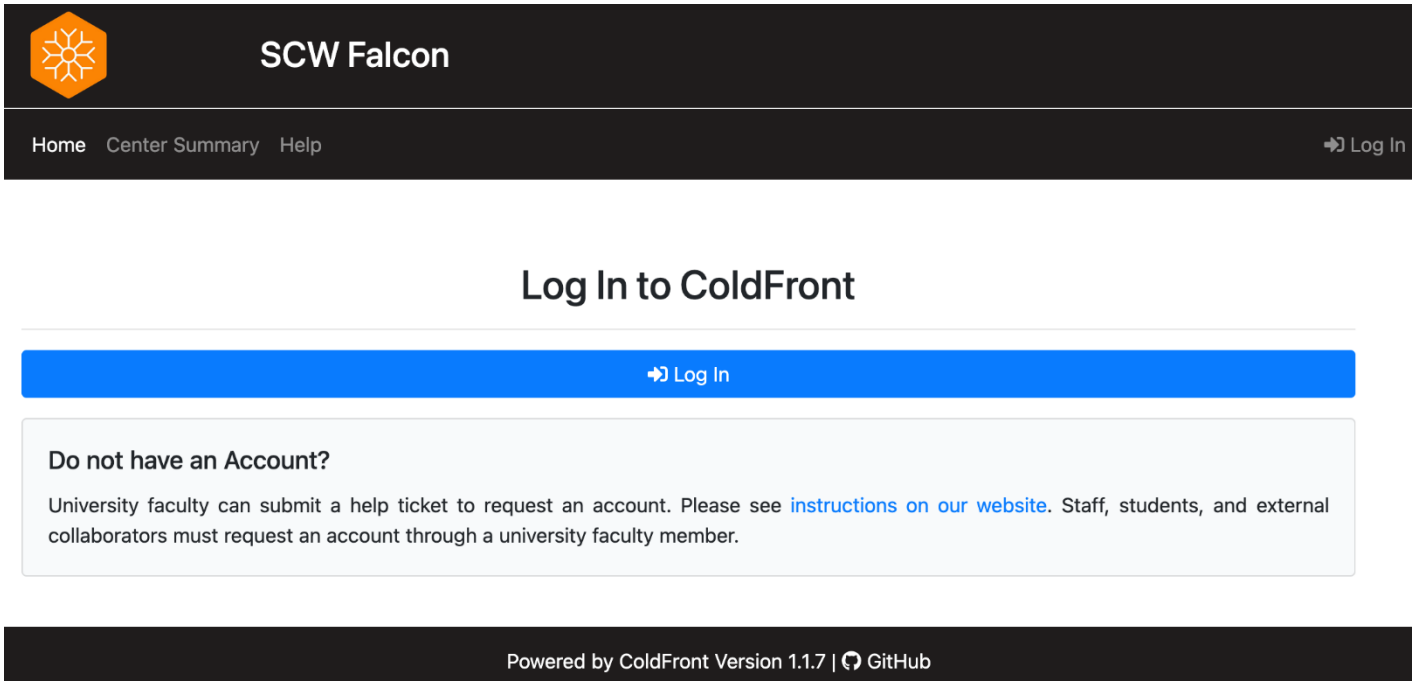


arcca New User Account III



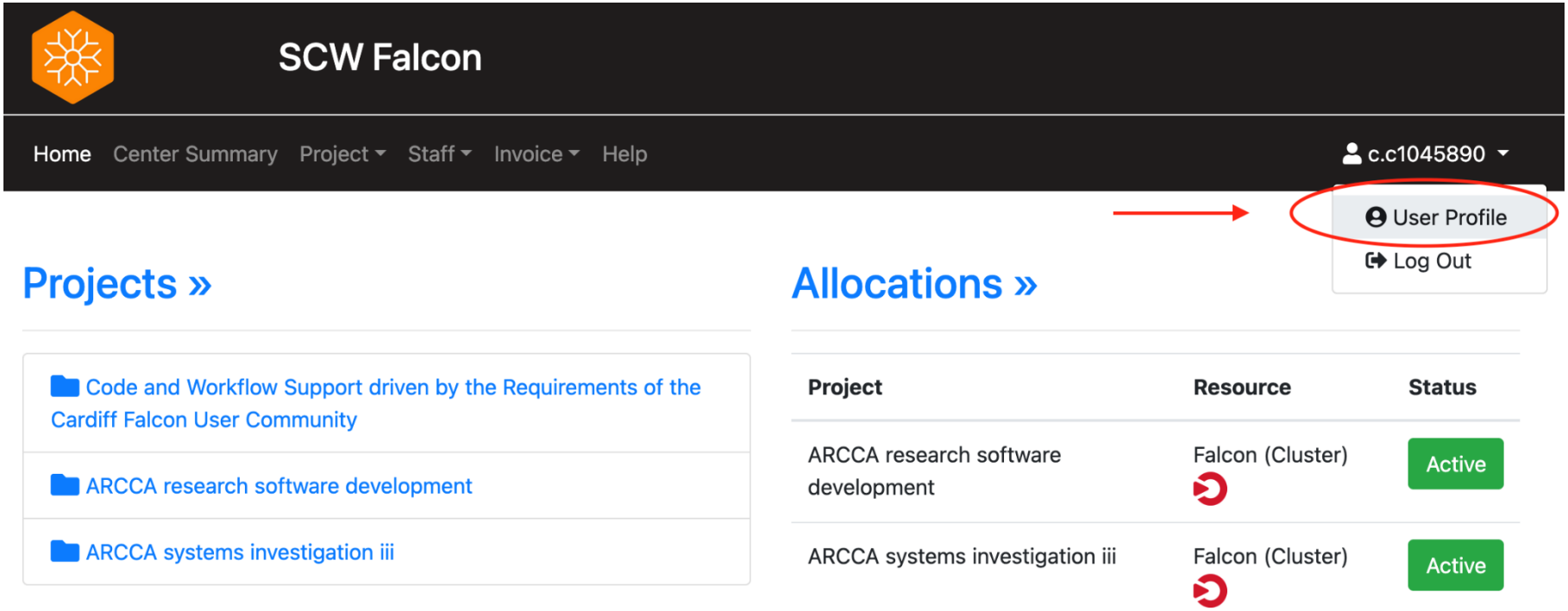
New Project Account - PI Driven

- After your user account is approved, visit: https://coldfront.cf.ac.uk/
- Login using your new Falcon username, typically ”c.” followed by your Cardiff username and the password setup in COGS.



New Project Account (PIs) - The Options.

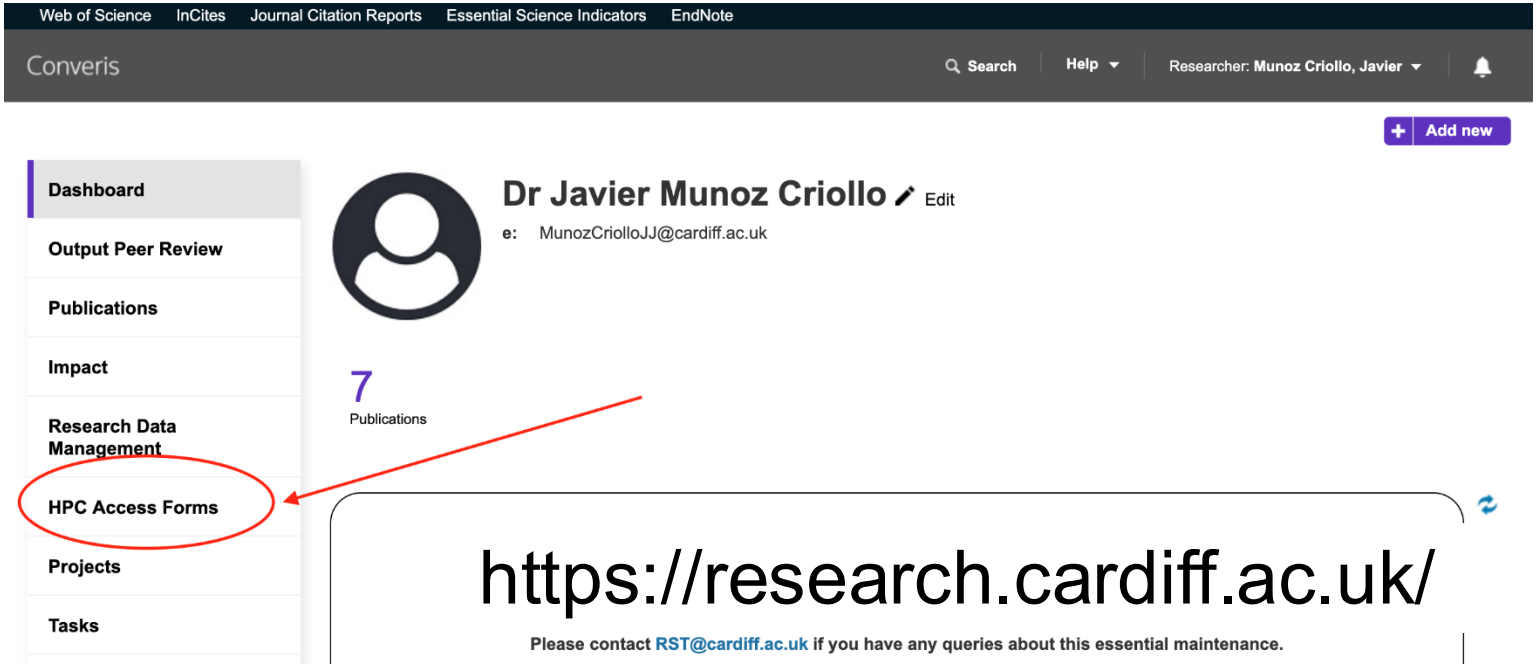
Option 1 — Quick Request from ColdFront portal



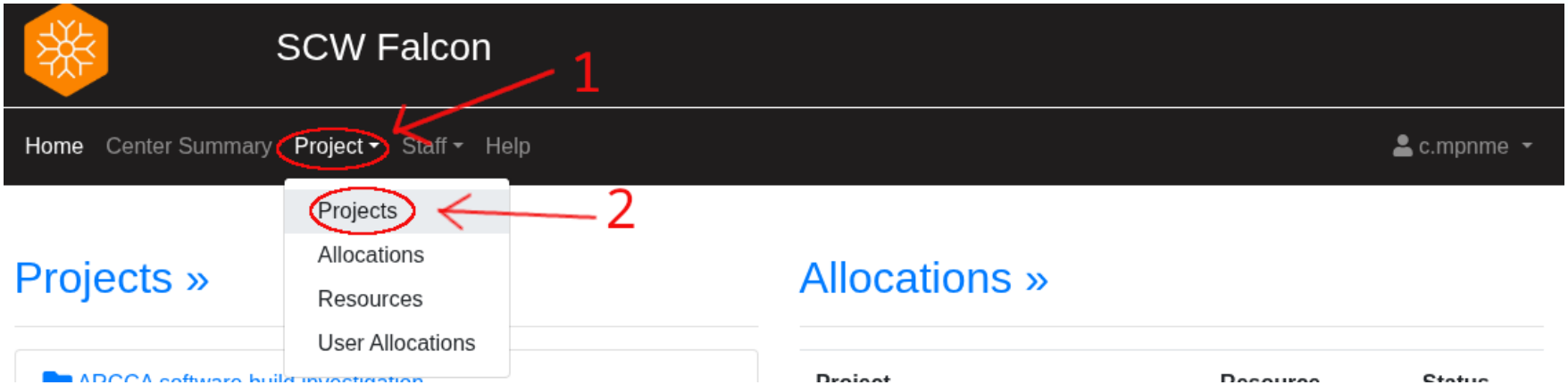
New Project Account (PIs) - The Options

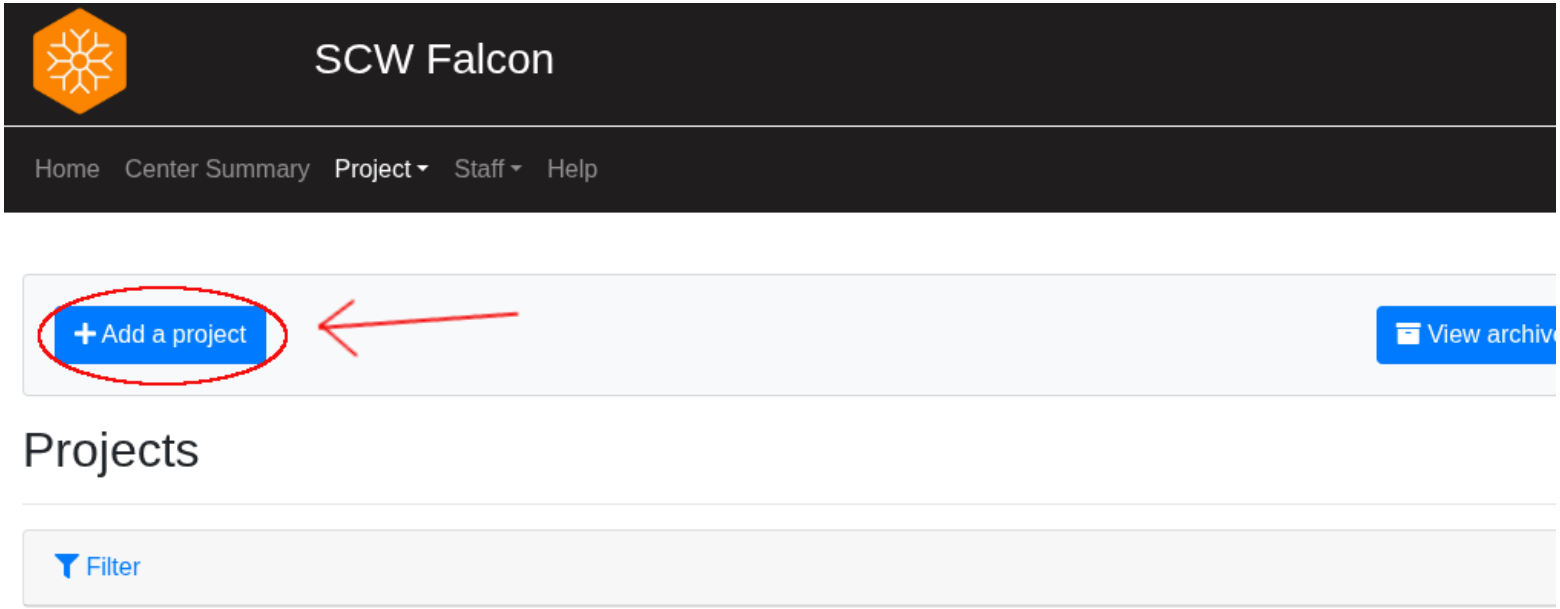
Option 2 — HPC Access Form (Research Portal)

- Complete and submit the HPC Access Form on the Research Portal at any time. An email will also be sent to you following a successful HPC Access Form submission.
- This also notifies ARCCA that you require PI status.

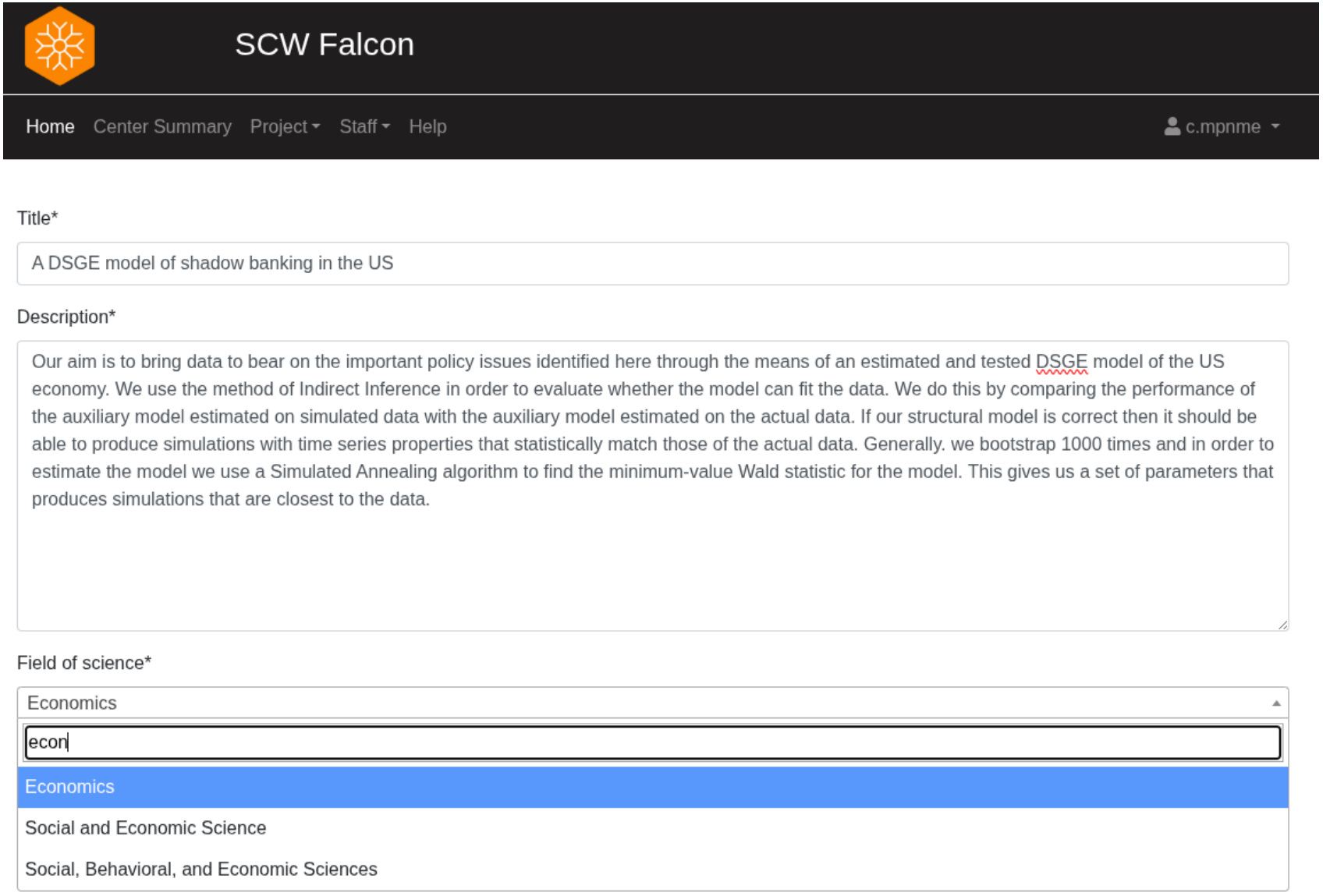


arcca New Project Account (PIs)

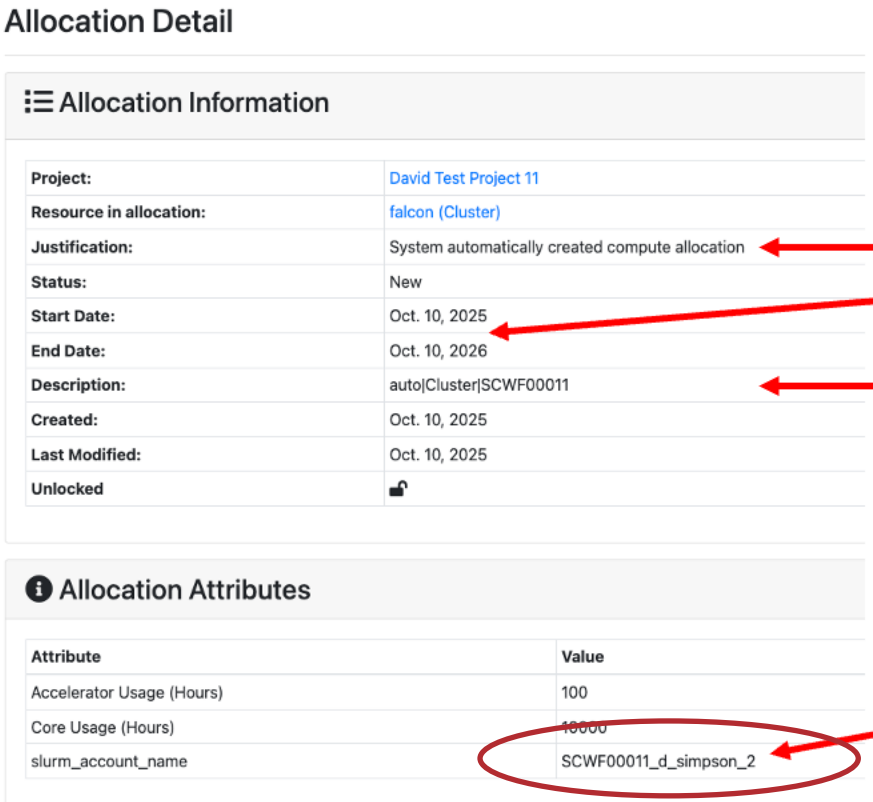




arcca New Project Account (PIs)



arcca New Project Account (PIs)



Pre-populated justification field, signifying automatic

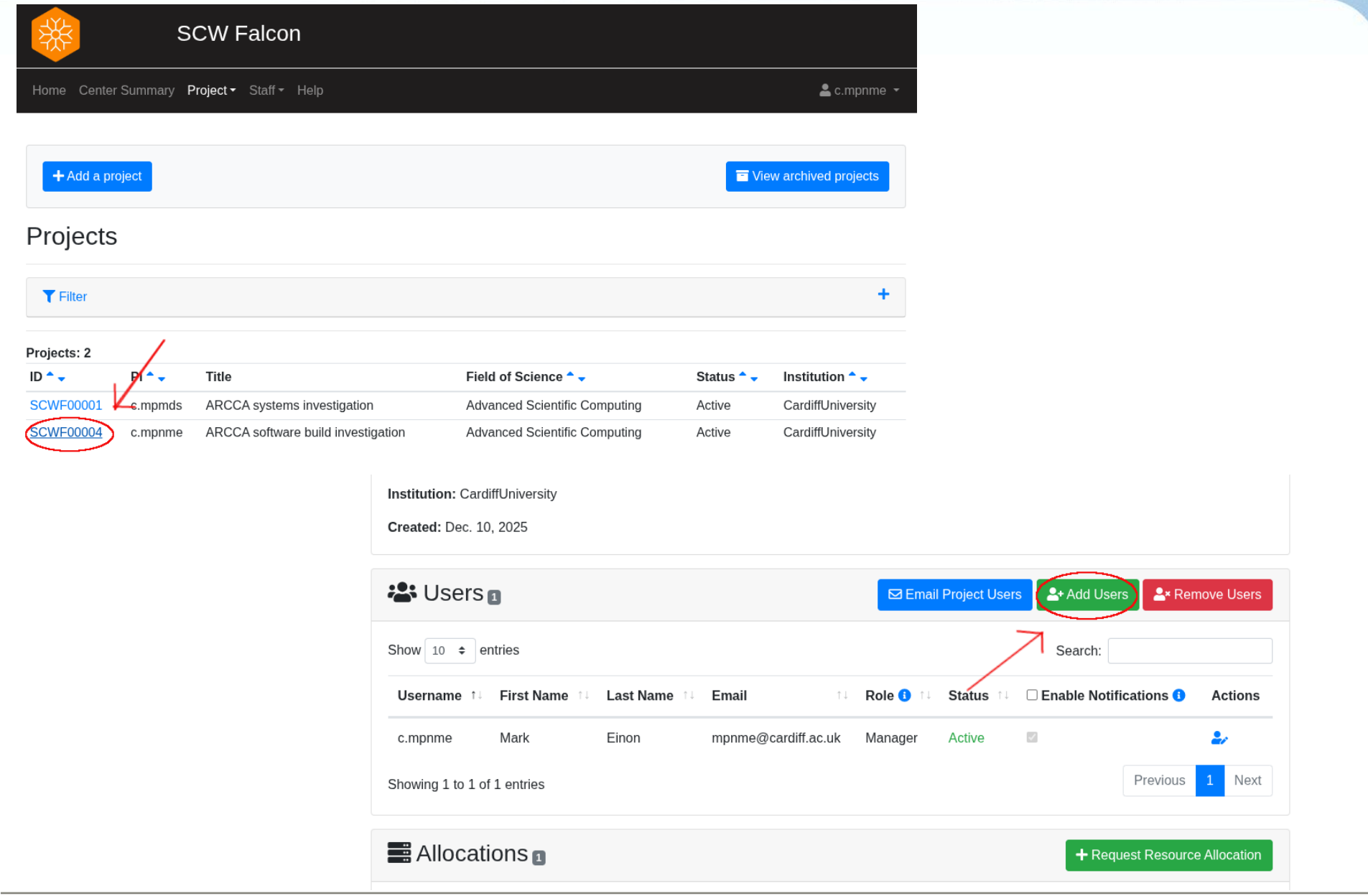
Validity period, aligns with 1 year project review

Pre-populated, may need changed in exceptional cases only

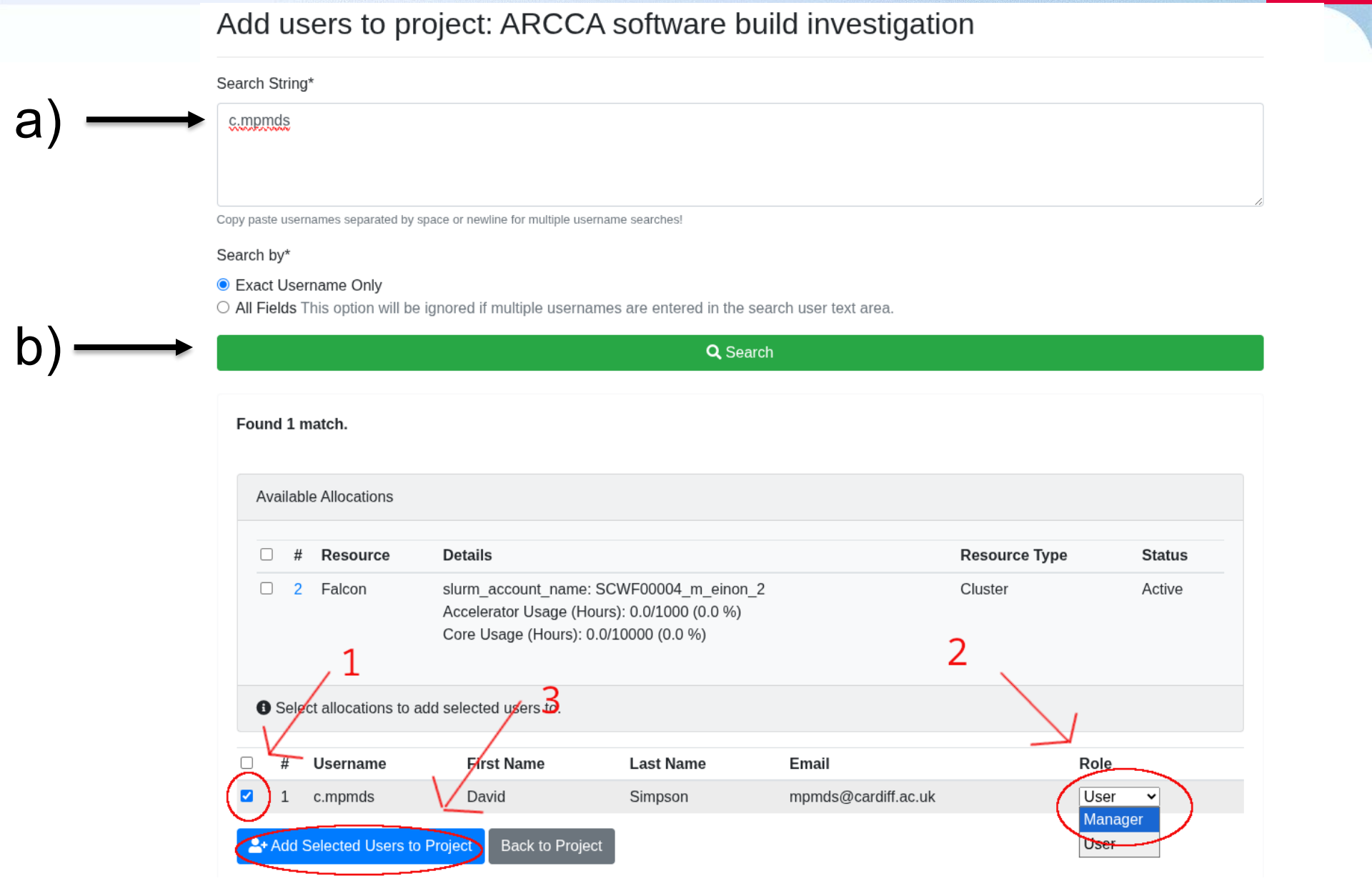
The slurm account name to use for job submission

format:
{projectcode}_{PIFirstInitial}_{PISurname}_{uniqueid}

Adding Users to a Project (PIs) I.

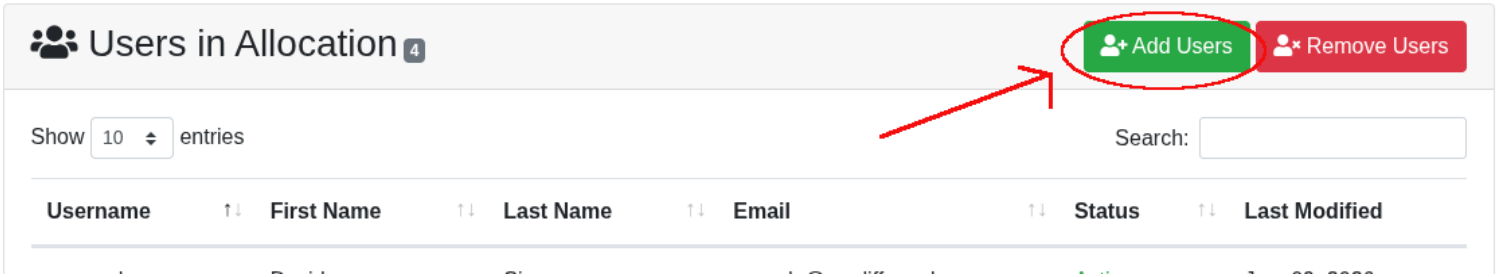


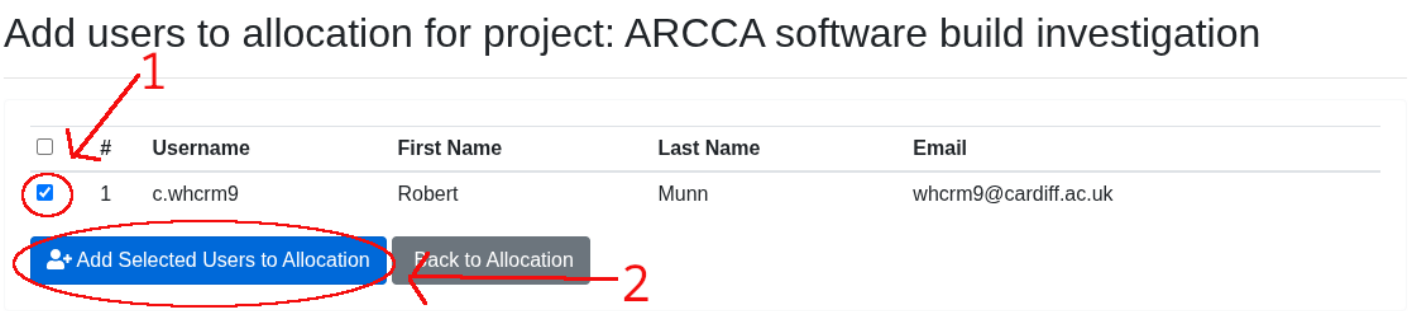
Adding Users to a Project (PIs) II



Adding users to a project's compute allocation (PIs)



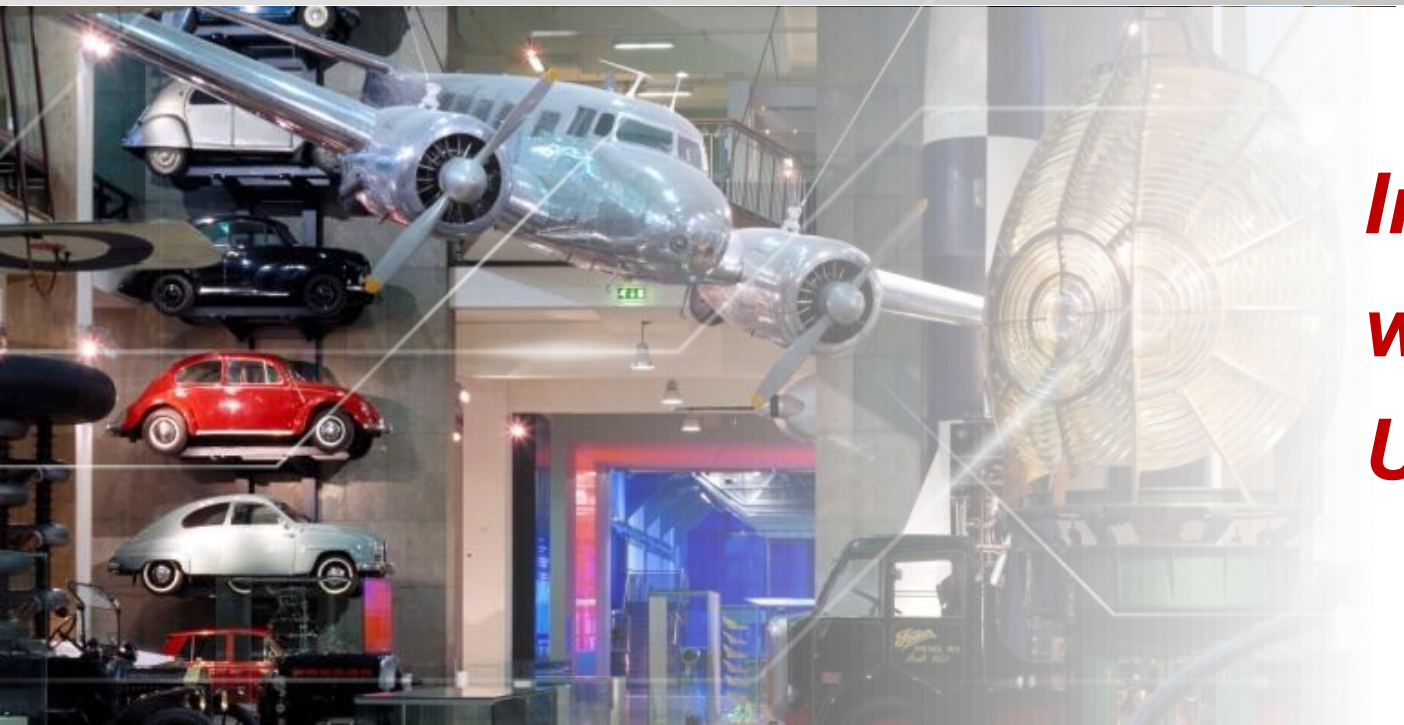




New Project Account (non-PIs)

- Once your user account is approved, your PI will be able to find you and add you to the right project. **That's all ☺**
- But – one final step - you will need to access **Coldfront** and read and accept a **software licensing agreement (EULA)** in order to submit jobs in the system after you have been added to a project.

Accessing Falcon and getting started



Interacting with and Using Falcon

How best to interact with and use Falcon

1. "Old-school" **command line** method. Still the ***most flexible and adequate for automation***. However**, this requires a knowledge of Linux, Bash and some SLURM commands.**

2. **Web-browser** based access (ARCCA OnDemand). Allows **easier access to interactive applications** (Rstudio, Jupyterlab, etc). **Main downside is queue wait times on a shared system.**

3. **Cloud** projects (AWS). Provides an opportunity to create virtual machines on an AWS platform (good option for Windows programs). This is **available for funded projects**.

Command Line

Use a terminal (natively available on Linux and Mac, Windows relies on third party apps).



Example using iTerm on Mac

On the command line users typically create ‘job scripts’, text files with instructions about how to run a job and resource requirements.

Command Line

Example job script: 384 cores on the Falcon "dev" partition to run a two-node, MPI application.

```
#!/bin/bashn
#SBATCH --ntasks=384
#SBATCH --ntasks-per-node=192
#SBATCH -o <output-file.txt>
#SBATCH -e <error-file.txt>
#SBATCH -J <job name>
#SBATCH --time=01:00:00
#SBATCH -partition=dev
#SBATCH --account=scwfXXXXX_j_doe_XX

module load compiler/intel/2018
module load mpi/intel/2018

mpirun -np 384 <code> <inputfiles>
```

Resource requirements and other metadata for the job

← User project account

Instructions to run the application (loading software, execution command)

Job Submission

Once your job script is prepared, it can be submitted to Slurm using the command “sbatch”:

sbatch <job-script-name.sh>

Alternatively, all the sbatch options can be provided interactively via the command line rather than in a job script:

```
sbatch --ntasks=384 --ntasks-per-node=192 --partition=compute --
account=--account=scwfXXXXX_j_doe_XX <run-script.sh>
```

For a full list of options, “**man sbatch**”

NOTE: It is recommended to set a maximum run time for the job:

```
#SBATCH time=8:00:00
```

Otherwise the job will be treated as requesting the maximum default time (likely increasing queue times).

Job Submission

- All jobs must be submitted via the batch scheduler – **Slurm**.
- **Jobs must NOT be run on the login nodes.**
- Slurm can be used to run parallel jobs across the system, or to reserve compute nodes to run interactively.
- The user policies and configuration of the queues are refined on an ongoing basis. Please use the default (**compute**) unless otherwise informed.
- At this stage, there is **a maximum run time** for jobs on the cluster of **72 hours (parallel).** This is "subject to negotiation" - **please be considerate to other users on the cluster**.

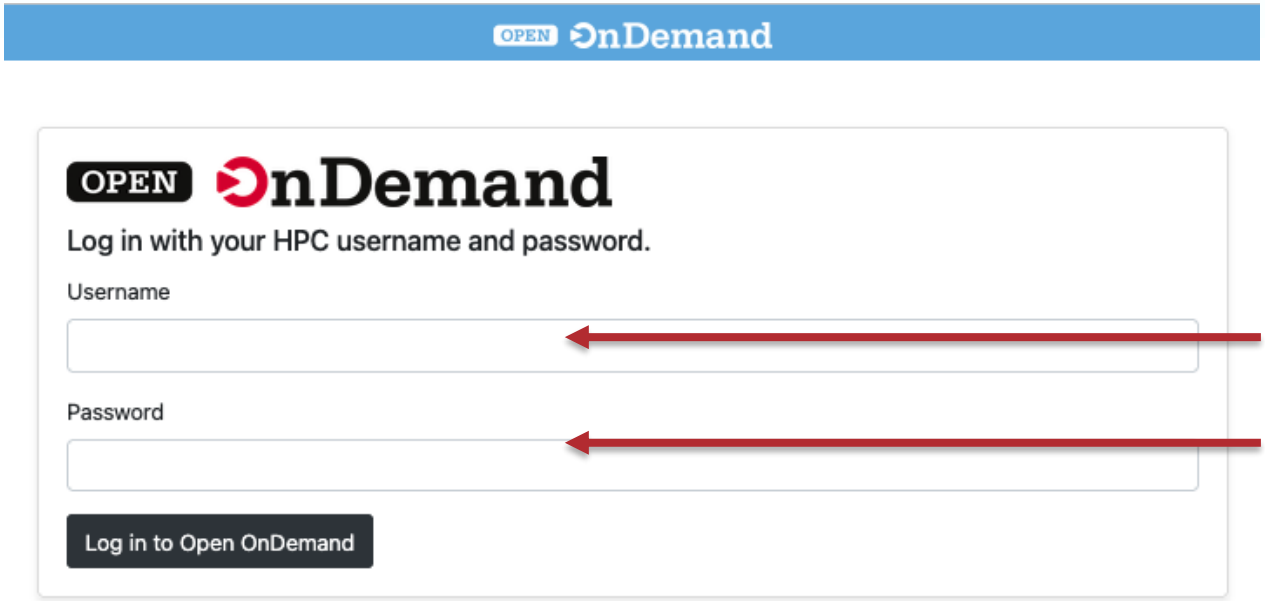
Feedback & suggestions welcomed on user policy & storage allocations (quotas)

arcca Open OnDemand

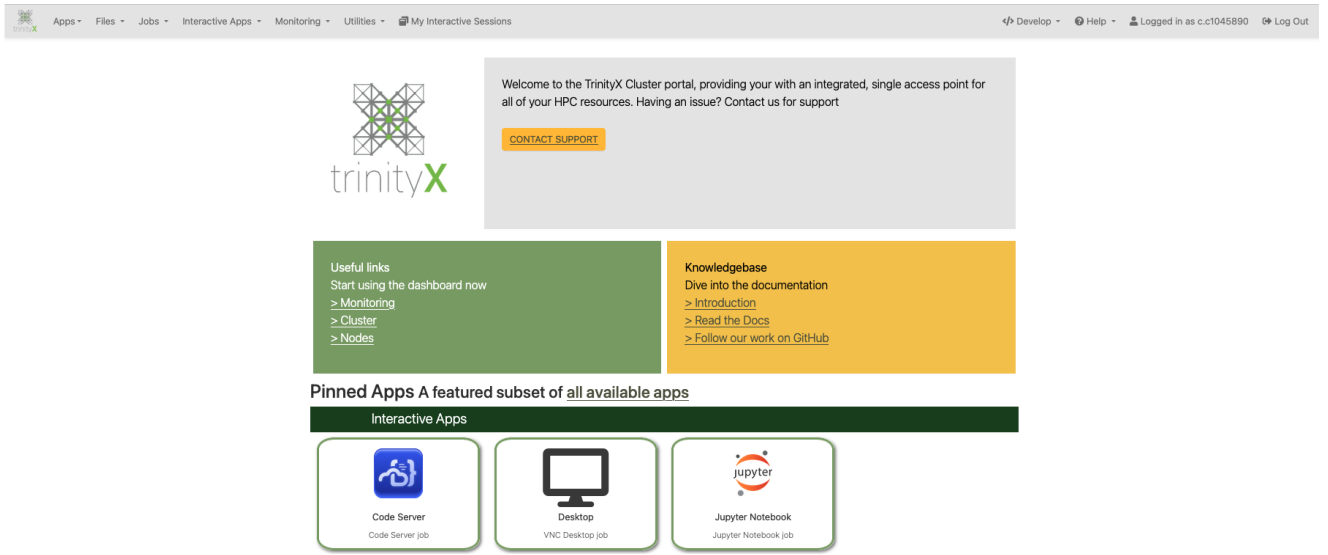
- Open OnDemand is an open-source, web-based portal designed to make **High-Performance Computing (HPC)** systems accessible and easy to use **via a web browser**.
- Its primary purpose is to allow researchers and students to manage files, submit jobs, and run interactive graphical applications remotely.
  - **Simplified HPC Access:** Provides a user-friendly interface that lowers the barrier to entry, removing the need for advanced Linux or command-line Knowle
  - **Interactive Application Access**: Enables running graphical applications (VS Code, RStudio, GUIs) directly on HPC clusters rather than just text-based batch jobs.dge.
  - **Web-Based File & Job Management:** Allows users to upload/download files, manage jobs, and monitor system resources in real-time.

https://falconlogin.cf.ac.uk:8080/

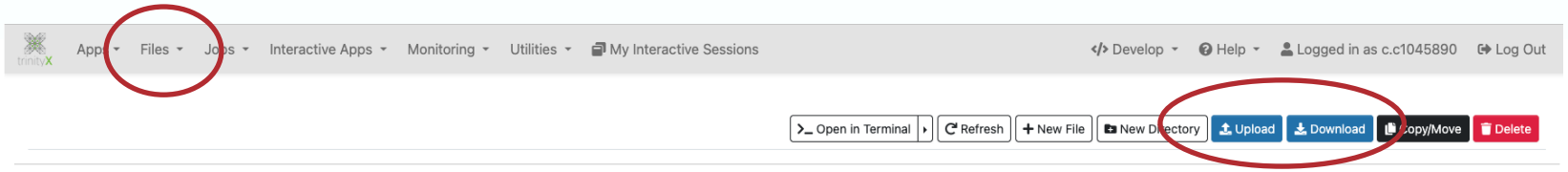
arcca Open OnDemand



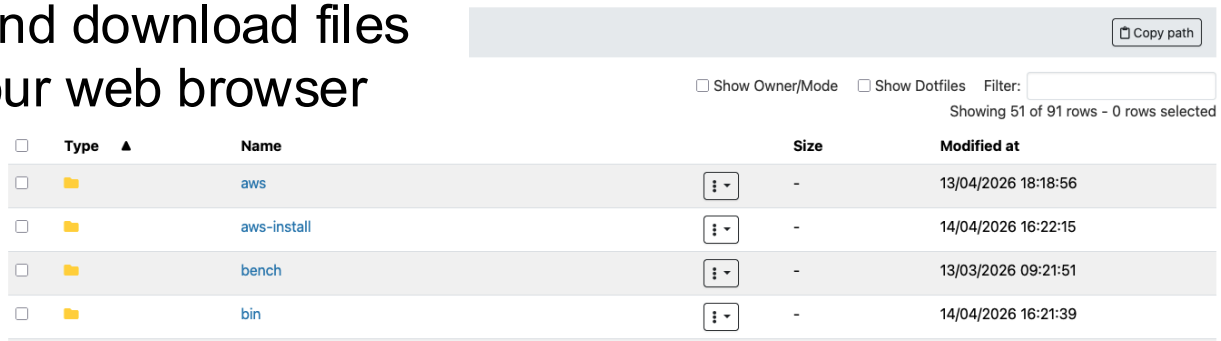
Use your Falcon username and password



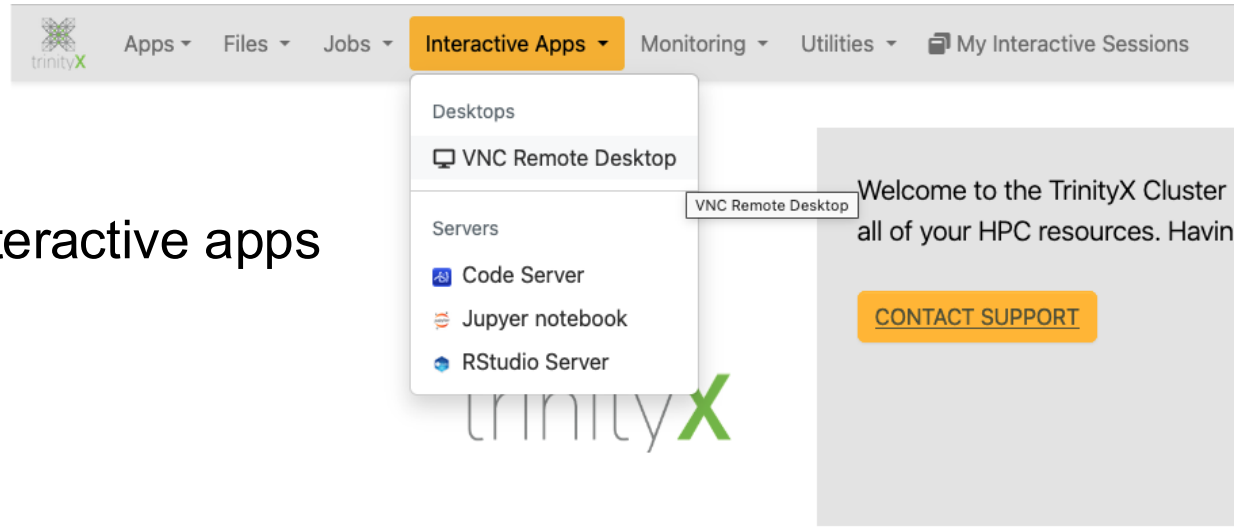
arcca Open OnDemand



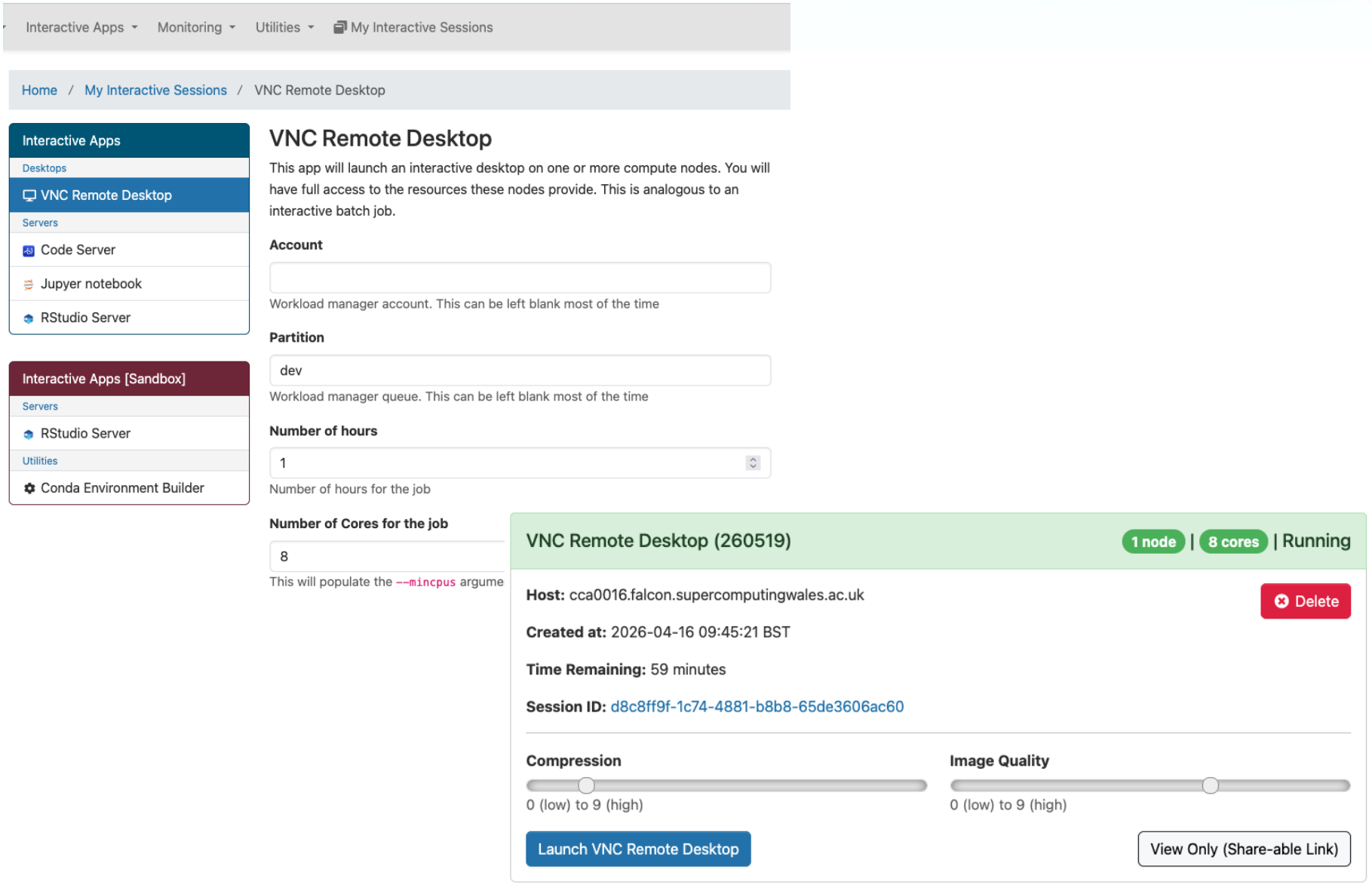
Upload and download files from your web browser



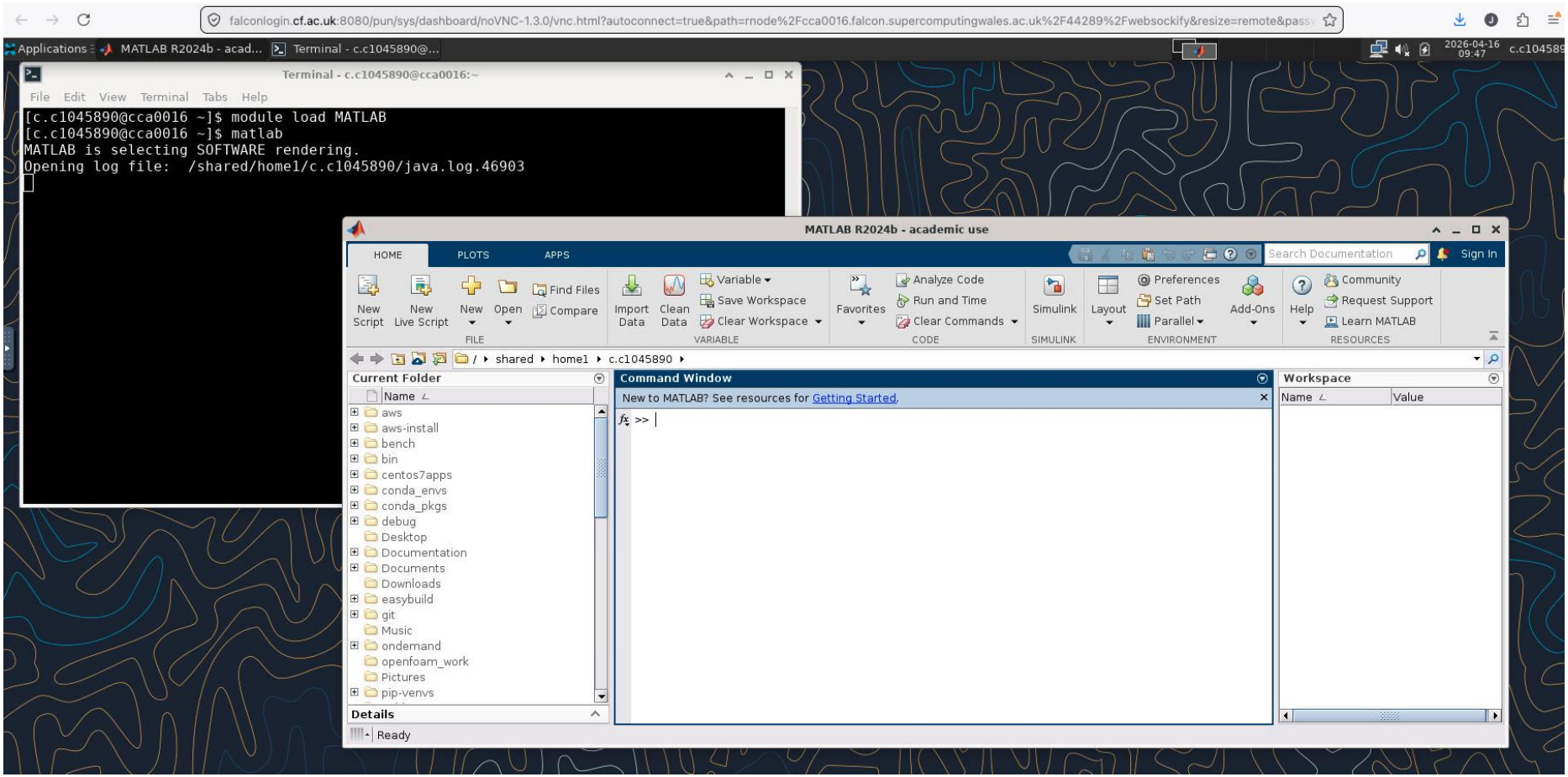
Access to interactive apps



arcca Open OnDemand



- Example of using MATLAB on Falcon's Open OnDemand VNC app.



Help and training

- For Cardiff users only, all Falcon related support queries: arcca-help@cardiff.ac.uk or IT-support@cardiff.ac.uk
- For general queries: arcca@cardiff.ac.uk
- For all Falcon help including FAQs, various guides and best practice information visit: https://wiki.arcca.cf.ac.uk/
- Subscribe to our community mailing list: https://mailman.cardiff.ac.uk/postorius/lists/arcca-community.mailman.cardiff.ac.uk/

ARCCA Docs (VPN required!)

ARCCA mailing list

arcca Survey – Please Complete!



https://app.onlinesurveys.jisc.ac.uk/s/cardiff/arcca-training-survey-2026-2027